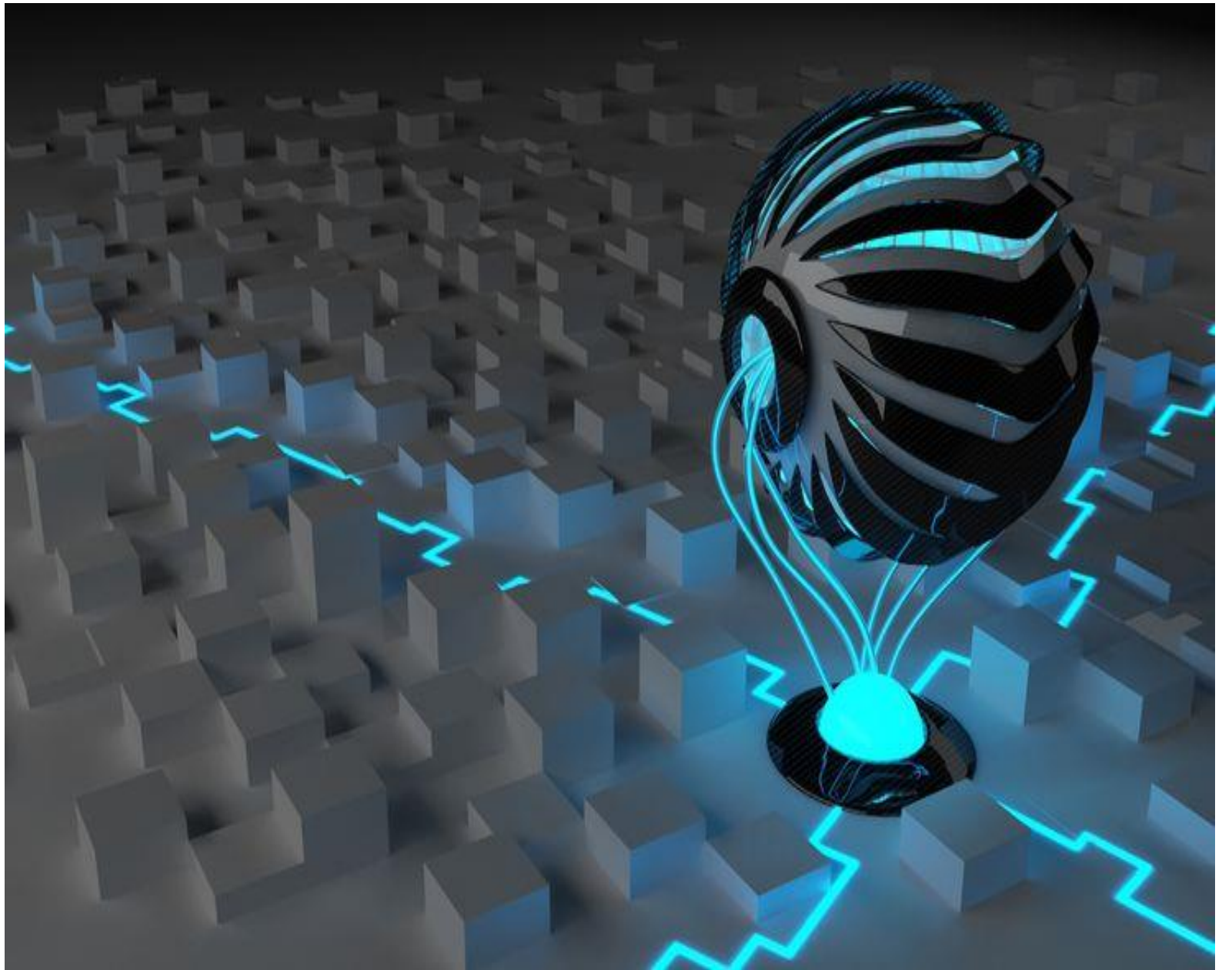


Computer Graphics

Chapter 3

3-Dimensional Geometric Transformation



Three –Dimensional Transformations

- Three –dimensional translation, rotation, scaling, reflection, shear transforms
- Three-dimensional composite transformation
- Three-dimensional viewing pipeline, world to screen viewing transformation
- Projection concepts (orthographic, parallel, perspective projections)

3-Dimension

- ❑ Three dimensional space is a geometric 3 parameters model of the physical universe in which all known matter exists
- ❑ These three dimensions can be labelled by a combination of length, breadth, and depth
- ❑ This third dimension allows for rotation and visualization from multiple perspectives
- ❑ Any three directions can be chosen, provided that they do not all lie in the same plane

3 Dimensional Object

- ❑ A three-dimensional (3D) object is an entity that occupies space in three dimensions: length, breadth (width), and depth (height)
- ❑ **Volume**
 - ❑ A 3D object has volume, which is the amount of space it occupies
- ❑ **Surfaces**
 - ❑ It is bounded by surfaces, which can be flat (planes) or curved.
- ❑ **Vertices, Edges, and Faces**
 - ❑ A 3D object typically has vertices (corners), edges (lines where two surfaces meet), and faces (flat or curved surfaces)
- ❑ **Perspective**
 - ❑ 3D objects can be viewed from different angles, providing various perspectives.
- ❑ **Types of objects** : Geometrical shapes Cube, Sphere, trees, terrains, clouds, rocks, glass, hair, furniture, human body, etc

3-Dimension graphics

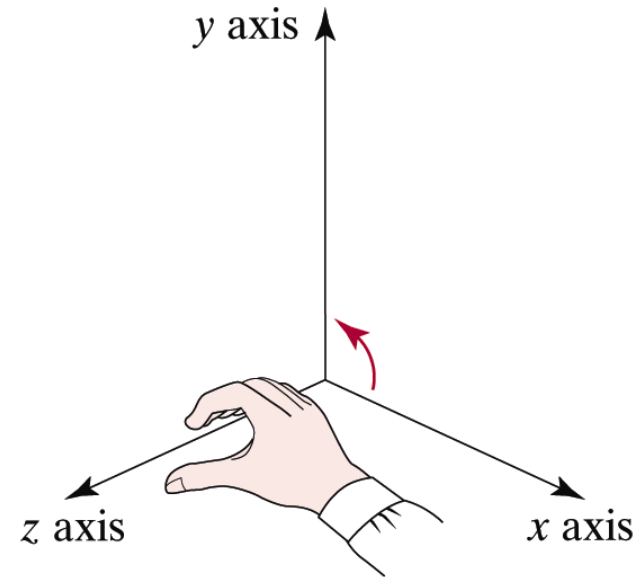
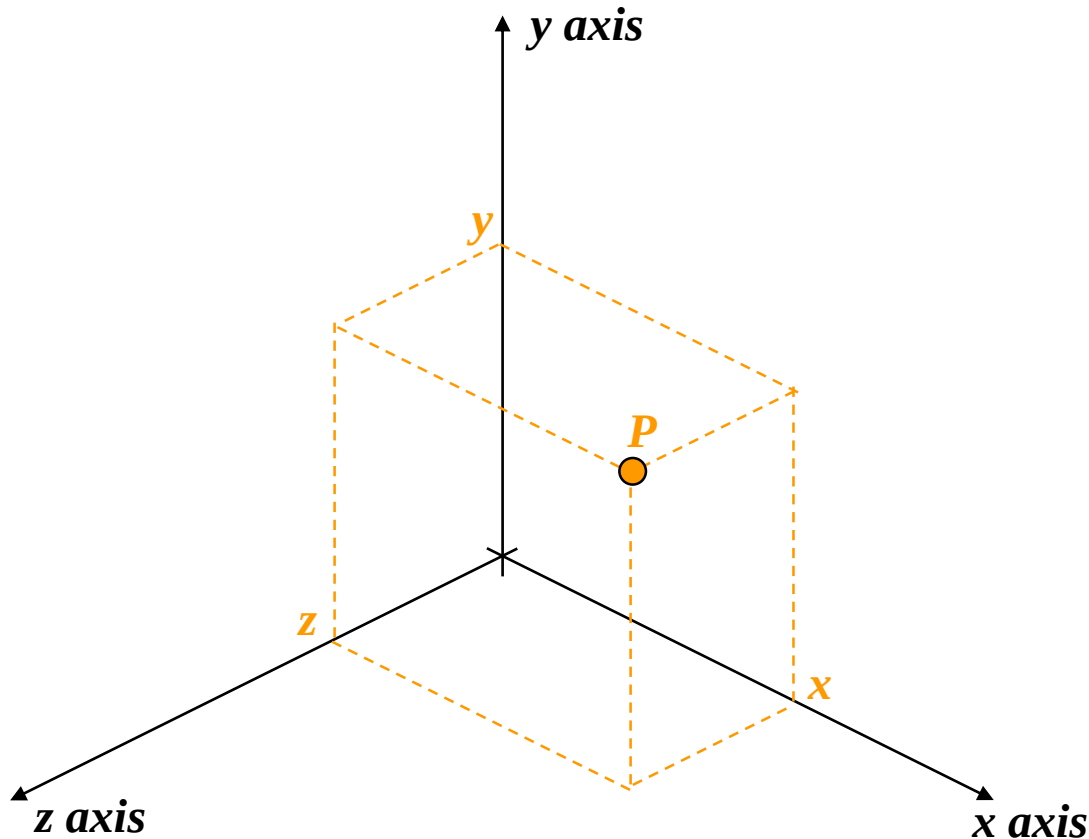
- ❑ 3D computer graphics use a three-dimensional representation of geometric data
- ❑ This data is stored in a computer for calculations and rendering 2D images
- ❑ 3D computer graphics involve creating and displaying images that simulate three-dimensional space, offering depth and realism beyond two-dimensional images
- ❑ Widely used in video games, movies, architecture, virtual reality (VR), augmented reality (AR), and various simulations
- ❑ 3D graphics enhance visual experiences and aid in design and visualization tasks

3-D Geometric and Modeling Transformations

- ❑ Just as 2D transformation can be represented by 3 x 3 matrices using homogeneous coordinate, 3D can be represented by 4 x 4 matrices
- ❑ Homogenous coordinate representation of point in 3D space is given by
 - ❑ (x_h, y_h, z_h, h)
 - ❑ $x = x_h / h$
 - ❑ $y = y_h / h$
 - ❑ $z = z_h / h$
- ❑ Therefore, (x, y, z) in Cartesian system is represented as $(x, y, z, 1)$ in homogeneous co-ordinate system

3D Coordinate System

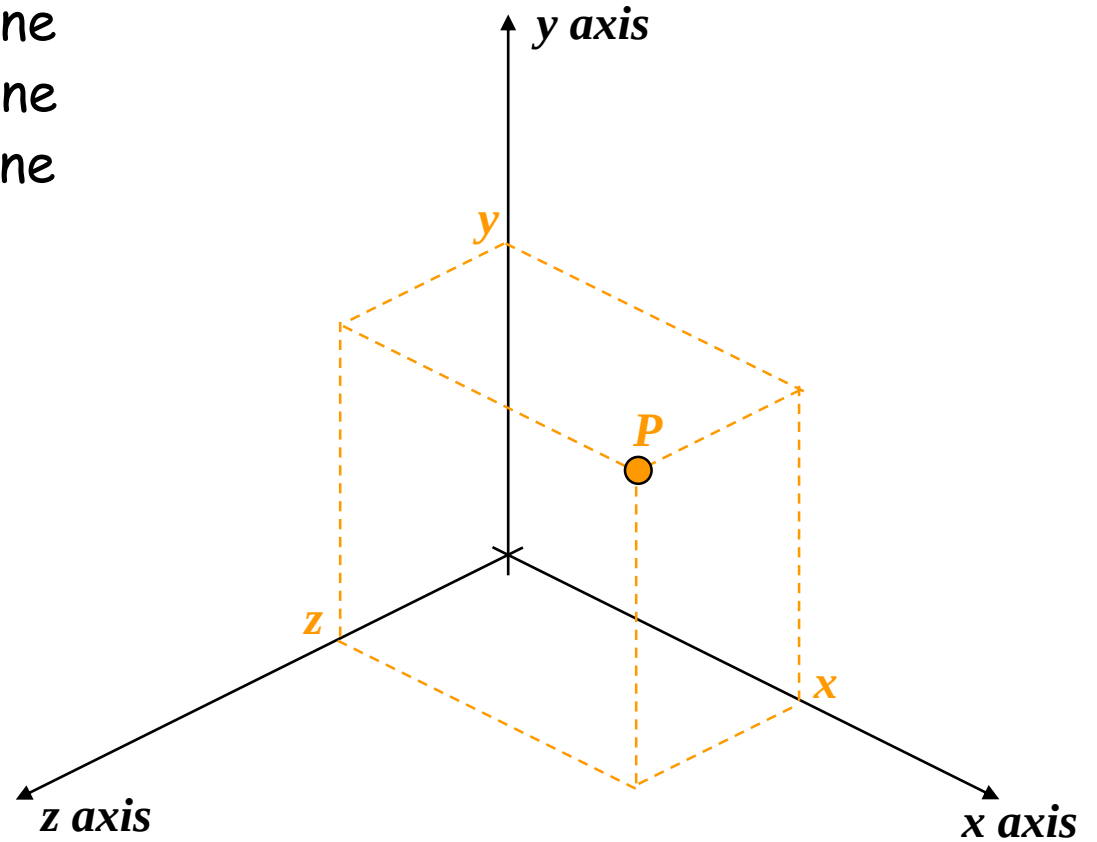
- 3D coordinate system consists of a reference point called origin and three mutually perpendicular passing through origin.



Right-Hand
Reference System

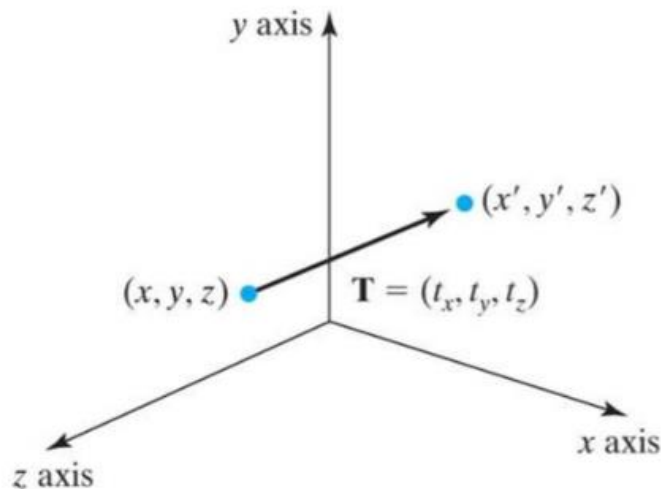
□ A point $P(x,y,z)$ specifies that point is

- At a distance of x units from YZ -plane
- At a distance of y units from XZ -plane
- At a distance of z units from XY -plane

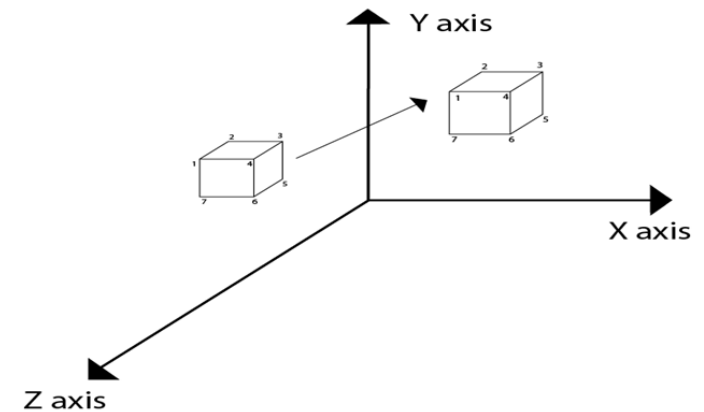


3D Translation

- ❑ Parameter t_x , t_y , t_z specify transition distances for the coordinate directions x , y , z .
- ❑ A point is translated from position $P(x, y, z)$ to position $P'(x', y', z')$ with the matrix operation as:



$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



- This matrix representation is equivalent to three equations:

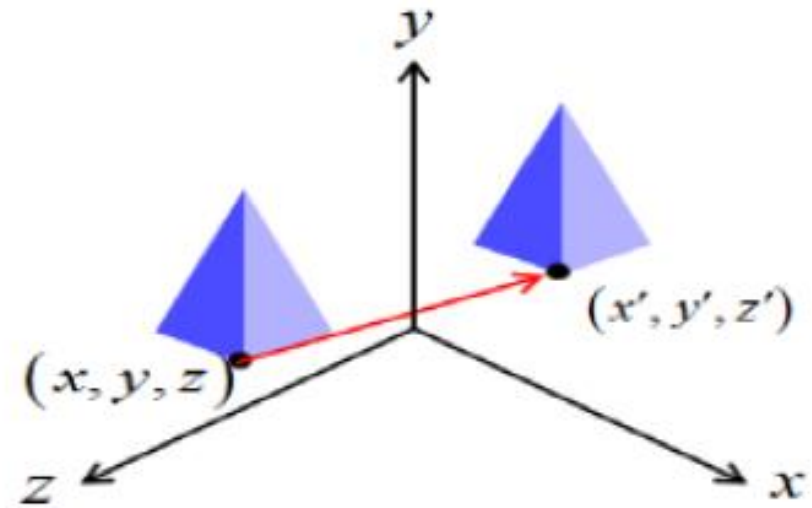
$$x' = x + t_x$$

$$y' = y + t_y$$

$$z' = z + t_z$$

$$\mathbf{P}' = \mathbf{T} \cdot \mathbf{P}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



3-D Scaling

□ Scaling about origin

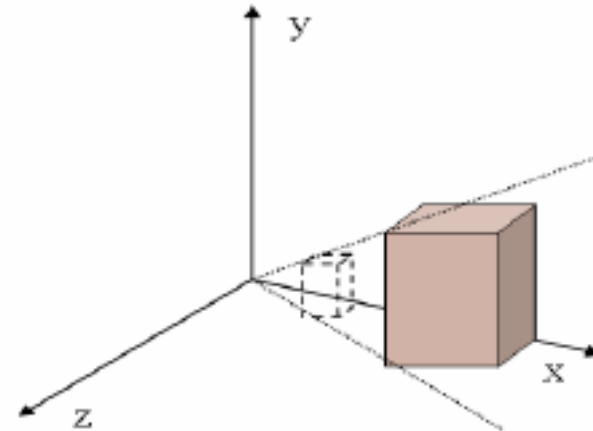
□ Matrix representation for scaling transformation of a position $P = (x, y, z)$ relative to the coordinate origin can be written as;

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$x' = S_x \cdot x$$

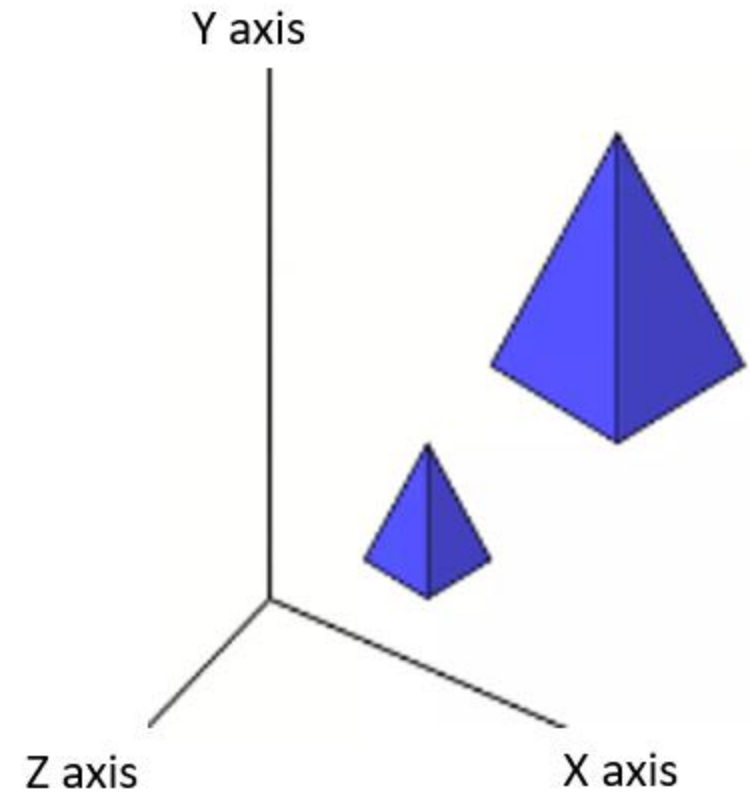
$$y' = S_y \cdot y$$

$$z' = S_z \cdot z$$



Fixed Point Scaling

- For scaling of point $P(x, y, z)$ w.r.t to fixed point (x_f, y_f, z_f) can be represented with the following transformation.
1. Translate the fixed point to the origin.
 $T(-x_f, -y_f, -z_f)$
 2. Apply scaling w.r.to origin. $S(s_x, s_y, s_z)$
 3. Translate the fixed point back to its original position. $T(x_f, y_f, z_f)$

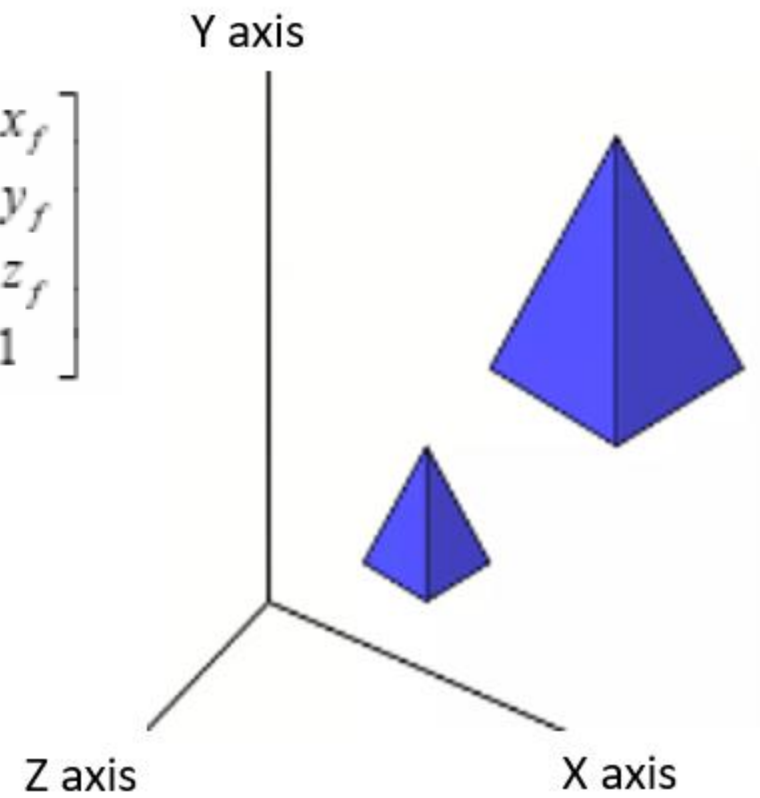


- Fixed Point Scaling

$$S(x_f, y_f, z_f, s_x, s_y, s_z) = T(x_f, y_f, z_f) \cdot S(s_x, s_y, s_z) \cdot T(-x_f, -y_f, -z_f)$$

$$= \begin{bmatrix} 1 & 0 & 0 & x_f \\ 0 & 1 & 0 & y_f \\ 0 & 0 & 1 & z_f \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & -x_f \\ 0 & 1 & 0 & -y_f \\ 0 & 0 & 1 & -z_f \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 & (1-s_x)x_f \\ 0 & s_y & 0 & (1-s_y)y_f \\ 0 & 0 & s_z & (1-s_z)z_f \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



3D Reflection

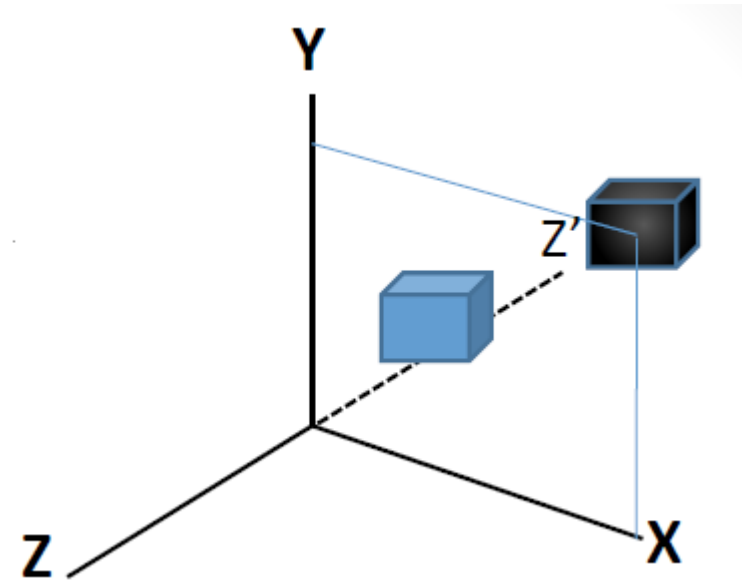
- In 3D-reflection the reflection takes place about a plane
- **Reflection in xy plane (z-axis)**
- This transformation changes the sign of the z coordinates, leaving the x and y coordinate values unchanged.

$$x' = x$$

$$y' = y$$

$$z' = -z$$

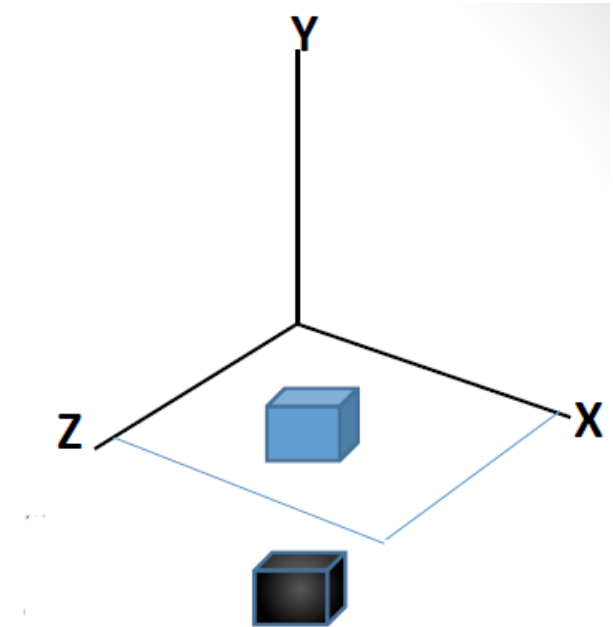
$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



Reflection in xz plane (*y*-axis)

- This transformation changes the sign of the z coordinates, leaving the x and y coordinate values unchanged
- $x' = x$
- $y' = -y$
- $z' = z$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



Reflection in yz plane (*x*-axis)

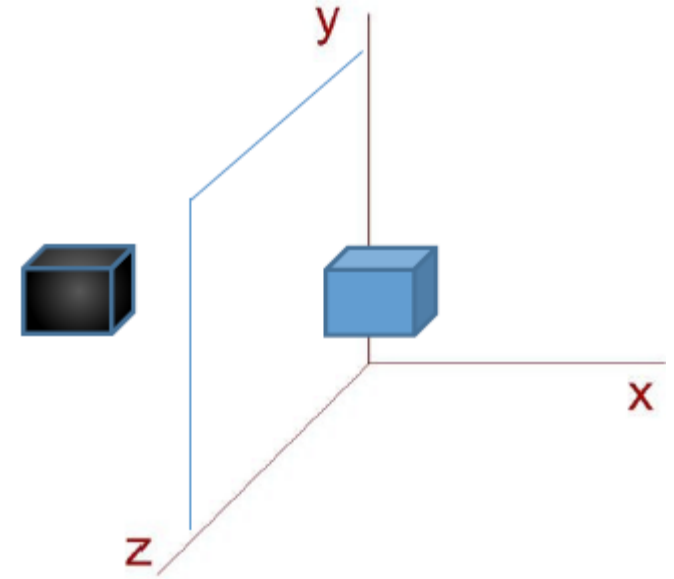
- This transformation changes the sign of the *x* coordinates, leaving the *y* and *z*-coordinate values unchanged

$$x' = -x$$

$$y' = y$$

$$z' = z$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



About any plane

Step-1: Translate the plane such that it passes through origin i.e. normal vector passes through origin

Step-2: Rotate the plane such that normal vector lies on one of the co-ordinate axis

Step-3: Perform reflection about the plane whose normal vector is one of the co-ordinate axis

Step-4: Rotate back the plane such that normal vector takes its original orientation

Step-5: Translate back the plane to its original position

3D Shearing

- Shearing transformations are used to modify objects shape.

(a) Z-axis shearing

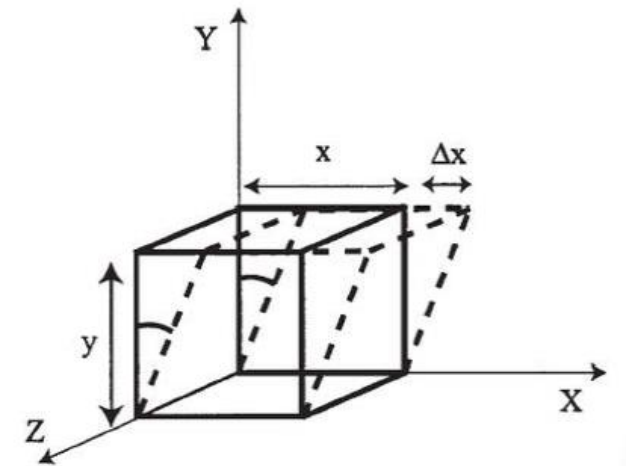
- This transformation alters x and y coordinates values by amount that is proportional to the z values while leaving z coordinate unchanged.

$$x' = x + s_{hx}Z$$

$$y' = y + s_{hy}Z$$

$$z' = z$$

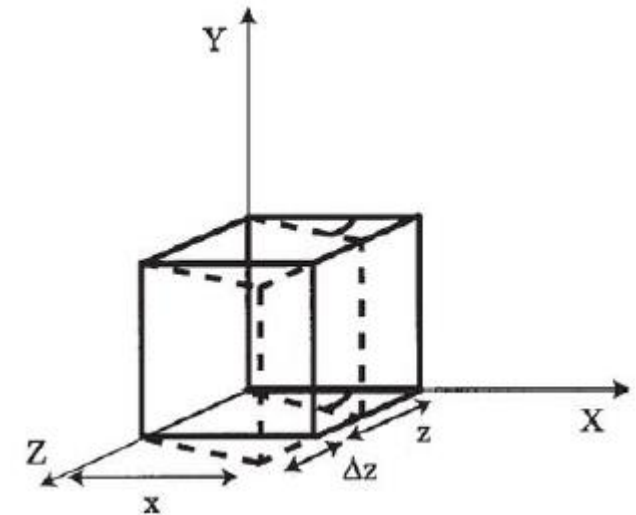
$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & s_{hx} & 0 \\ 0 & 1 & s_{hy} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



(b) X-axis shearing

- This transformation alters y and z coordinates values by amount that is proportional to the x values while leaving x- coordinate unchanged.

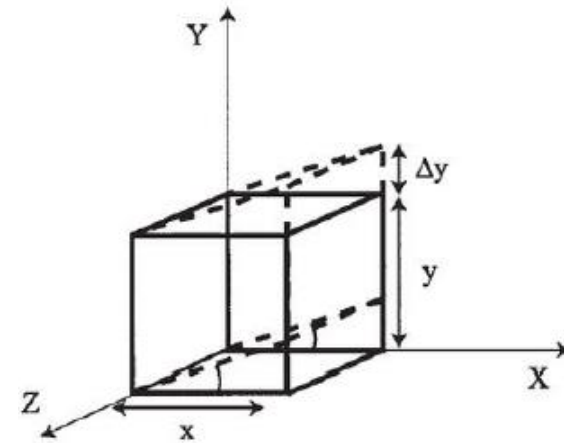
$$\begin{aligned}x' &= x \\y' &= y + s_{hy}x \\z' &= z + s_{hz}x\end{aligned} \quad \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ s_{hy} & 1 & 0 & 0 \\ s_{hz} & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



(c) Y- axis shearing

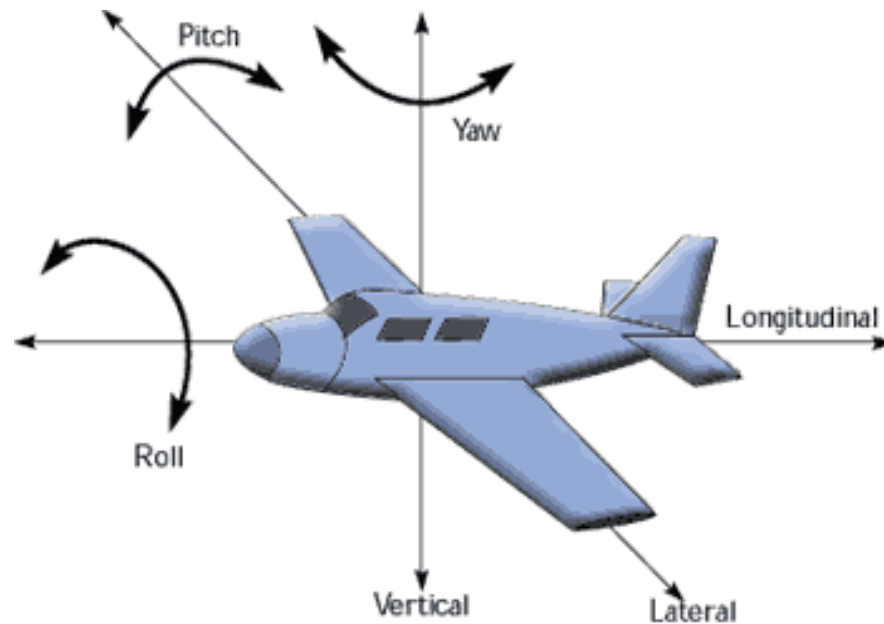
- This transformation alters x and z coordinates values by amount that is proportional to the y values while leaving y- coordinate unchanged.

$$\begin{aligned}x' &= x + s_{hx}y \\y' &= y \\z' &= z + s_{hz}y\end{aligned}\quad \begin{bmatrix}x' \\ y' \\ z' \\ 1\end{bmatrix} = \begin{bmatrix}1 & s_{hx} & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & s_{hz} & 1 & 0 \\ 0 & 0 & 0 & 1\end{bmatrix} \begin{bmatrix}x \\ y \\ z \\ 1\end{bmatrix}$$

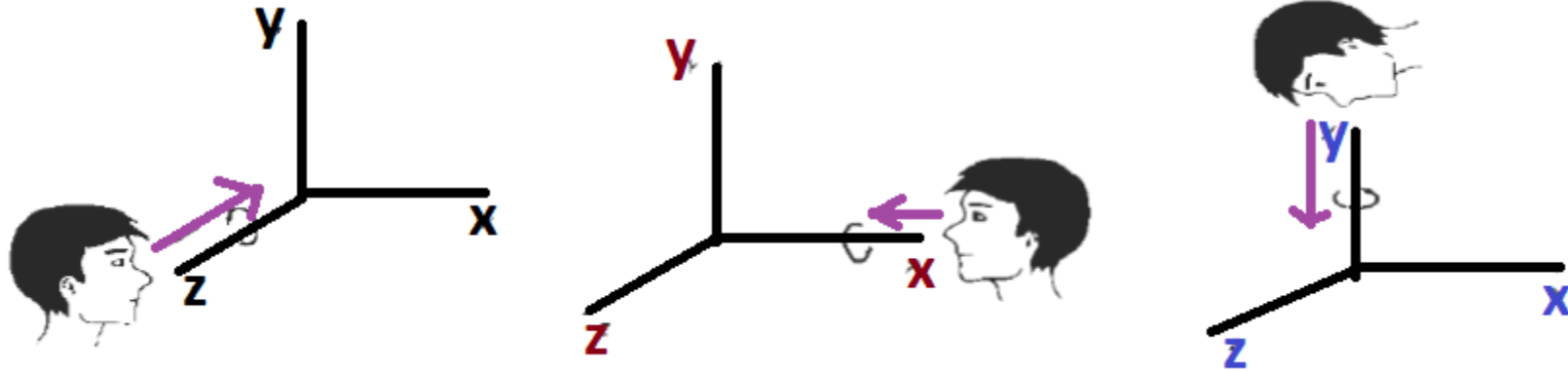


3D Rotation

- ❑ When we performed rotations in two dimensions we only had the choice of rotating about the z axis
- ❑ In the case of three dimensions we have more options
 - ❑ Rotate about x - pitch
 - ❑ Rotate about y - yaw
 - ❑ Rotate about z - roll



- Designate the axis of rotation about which the object is to be rotated and the amount of angular rotation



Coordinate axes Rotations:

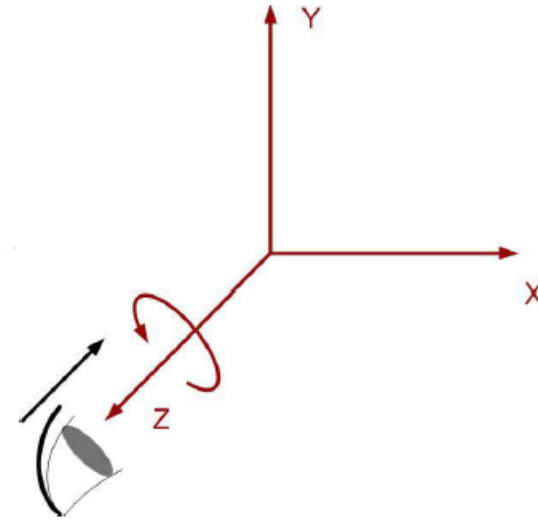
Rotation About z-axis:

□ Z-component does not change

$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

or,

$$P' = R_z(\theta) \cdot P$$



$$x' = x \cos \theta - y \sin \theta$$

$$y' = x \sin \theta + y \cos \theta$$

$$z' = z$$

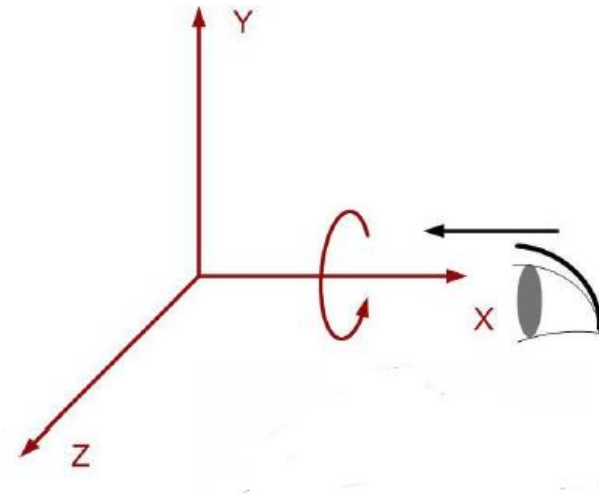
Coordinate axes Rotations:

Rotation About x-axis:

- *X-component does not change.*

$$P' = R_x(\theta) \cdot P$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



$$y' = y \cos \theta - z \sin \theta$$

$$z' = y \sin \theta + z \cos \theta$$

$$x' = x$$

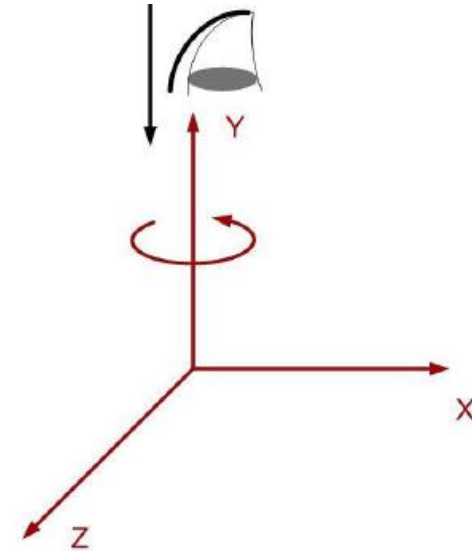
- **Coordinate axes Rotations:**

- Rotation About y-axis:

- *Y-component does not change.*

$$P' = R_y(\theta) \cdot P$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



$$z' = z \cos \theta - x \sin \theta$$

$$x' = z \sin \theta + x \cos \theta$$

$$y' = y$$

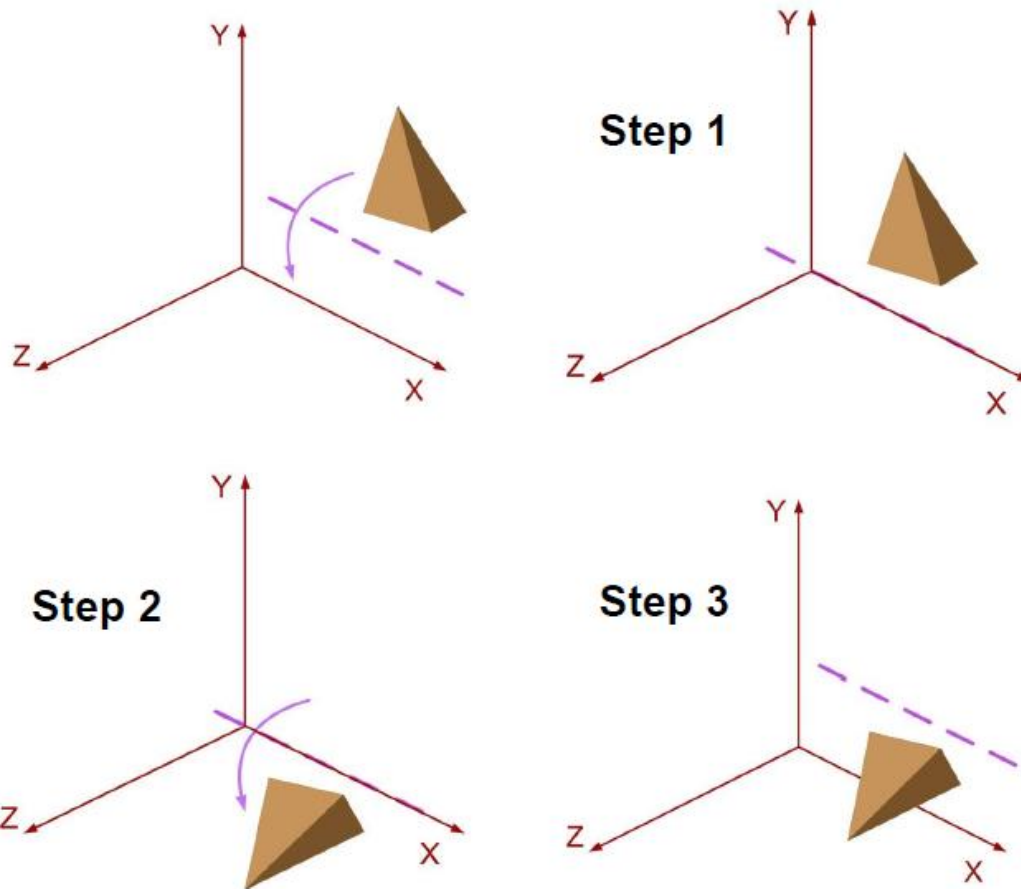
3D rotation about an axis that is parallel to any of the coordinate-axis

We need to perform some series of transformation:

- i. Translate object so that rotation axis coincides with the parallel coordinate axis
- ii. Perform specified rotation about that axis
- iii. Translate back the object so as to move rotation axis to original position

$$\mathbf{P}' = \mathbf{T}^{-1} \cdot \mathbf{R}_x(\theta) \cdot \mathbf{T} \cdot \mathbf{P}$$

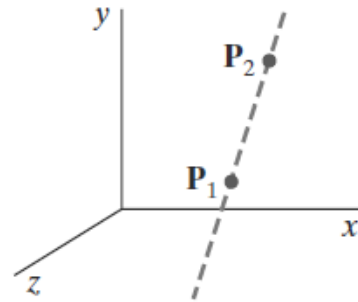
3D Rotation



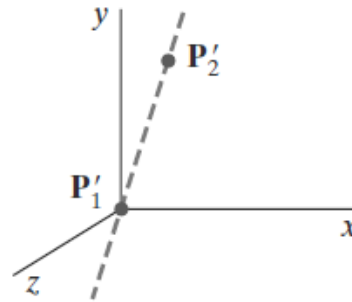
$$\mathbf{P}' = \mathbf{T}^{-1} \cdot \mathbf{R}_x(\theta) \cdot \mathbf{T} \cdot \mathbf{P}$$

Rotation about an Arbitrary Axis

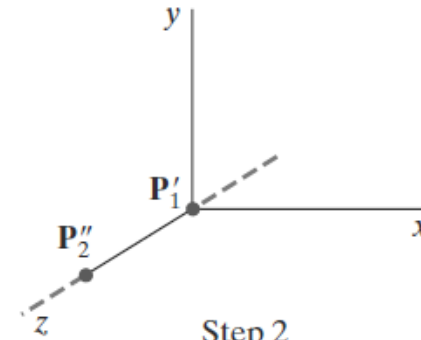
- When an object is to be rotated about an axis that is not parallel to one of the co-ordinate axes, we need to perform some series of transformation:
 1. **Translate** the object such that rotation axis passes through co-ordinate origin
 2. **Rotate** the object such that axis of rotation coincides with one of the co-ordinate axis
 3. Perform the specific **rotation** about selected coordinate axis
 4. Apply **inverse rotation** to bring the rotation axis back to its original orientation
 5. Apply **inverse translation** to bring the rotation axis back to its original position.



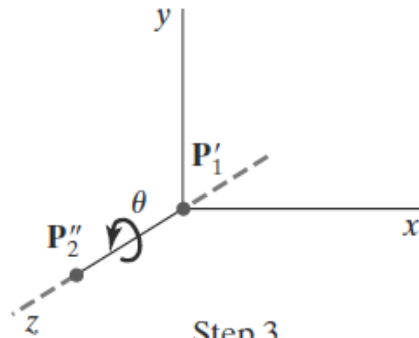
Initial
Position



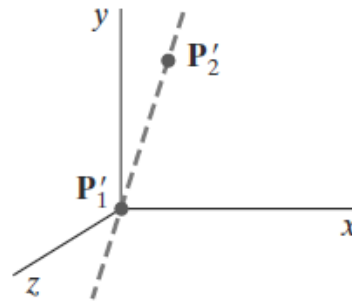
Step 1
Translate
 P_1 to the Origin



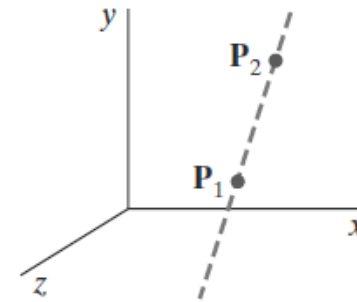
Step 2
Rotate P'_2
onto the z Axis



Step 3
Rotate the
Object Around the
 z Axis



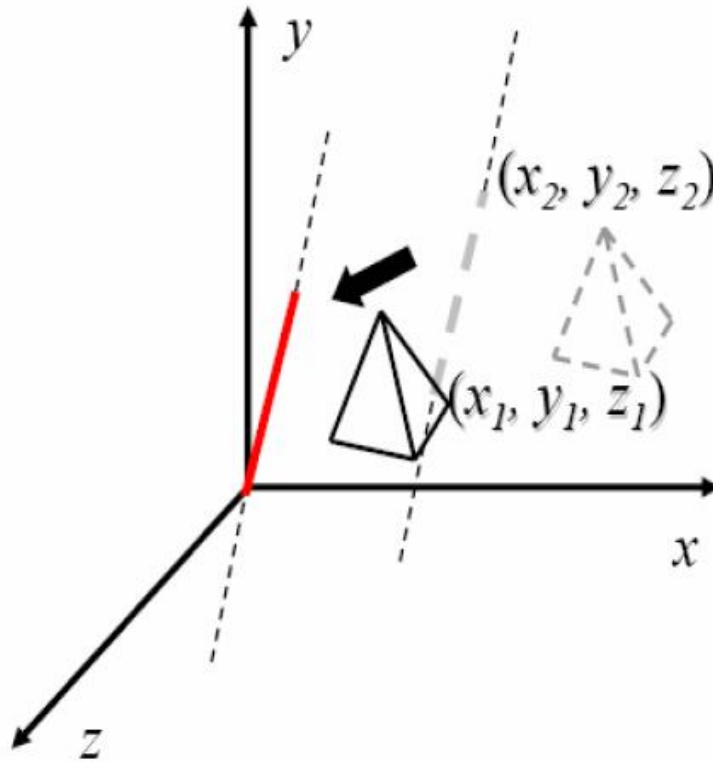
Step 4
Rotate the Axis
to Its Original
Orientation



Step 5
Translate the
Rotation Axis
to Its Original
Position

Rotation about an Arbitrary Axis

1. Translate the object such that rotation axis passes through coordinate origin.



$$\mathbf{T} = \begin{bmatrix} 1 & 0 & 0 & -x_1 \\ 0 & 1 & 0 & -y_1 \\ 0 & 0 & 1 & -z_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2. Rotate the object such that axis of rotation coincides with one of the coordinate axis.

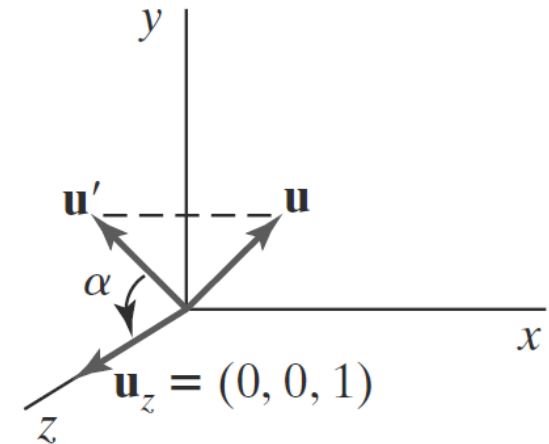
2.1 Rotate about x axis by α that brings point to the x-z plane

- This rotation angle is the angle between the projection of $\mathbf{u}$ in the yz plane and the positive z axis
- If we represent the projection of $\mathbf{u}$ in the yz plane as the vector $\mathbf{u}' = (0, b, c)$, then the cosine of the rotation angle α can be determined from the dot product of $\mathbf{u}'$ and the unit vector $\mathbf{u}_z$ along the z axis:

$$\cos \alpha = \frac{\mathbf{u}' \cdot \mathbf{u}_z}{|\mathbf{u}'| |\mathbf{u}_z|} = \frac{c}{d}$$

where d is the magnitude of $\mathbf{u}$:

$$d = \sqrt{b^2 + c^2}$$



2.1 Rotate about x axis by α that brings point to the x-z plane

- Similarly, we can determine the sine of α from the cross-product of $\mathbf{u}$ and $\mathbf{u}_z$. The coordinate-independent form of this cross-product is

and the Cartesian form for the cross-product gives us

$$\mathbf{u}' \times \mathbf{u}_z = \mathbf{u}_x |\mathbf{u}'| |\mathbf{u}_z| \sin \alpha$$

- Equating above two equations and noting that $|\mathbf{u}_z| = 1$ and $|\mathbf{u}| = d$, we get

$$\mathbf{u}' \times \mathbf{u}_z = \mathbf{u}_x \cdot b$$

$$d \sin \alpha = b$$

$$\sin \alpha = \frac{b}{d}$$

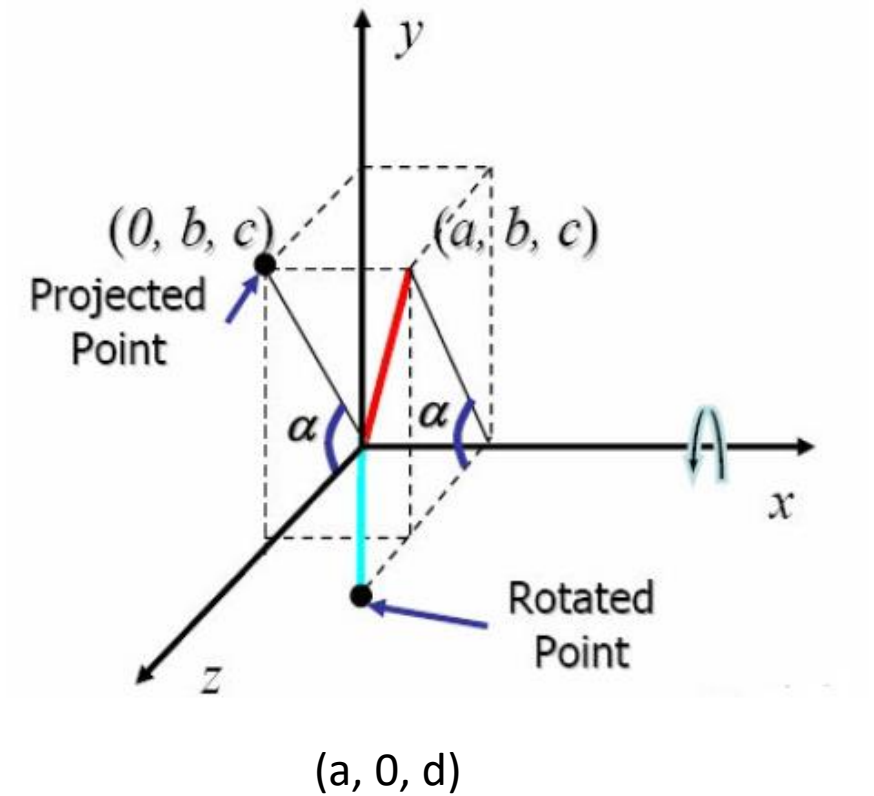
2.1 Rotate about x axis by α that brings point to the x-z plane

$$\sin \alpha = \frac{b}{\sqrt{b^2 + c^2}} = \frac{b}{d}$$

$$\cos \alpha = \frac{c}{\sqrt{b^2 + c^2}} = \frac{c}{d}$$

Rotation of this vector about the x axis and into the xz plane

$$\mathbf{R}_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha & 0 \\ 0 & \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c/d & -b/d & 0 \\ 0 & b/d & c/d & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



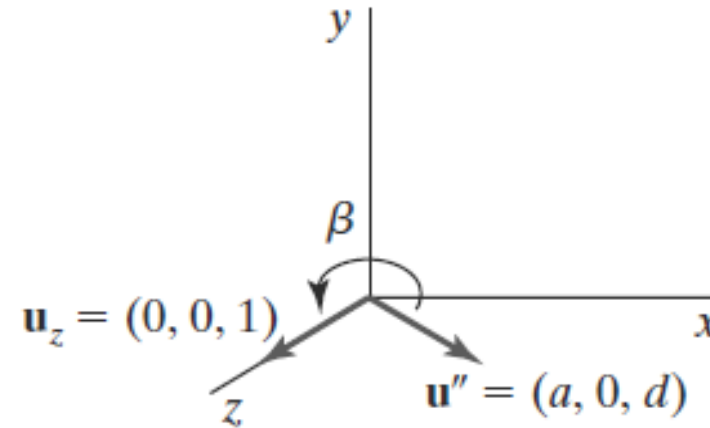
2.2) Rotate about y axis by β (clockwise) which brings point to the z-axis

- Positive rotation angle β aligns $\mathbf{u}''$ with vector $\mathbf{u}_z$
- we can determine the cosine of rotation angle β from the dot product of unit vectors $\mathbf{u}''$ and $\mathbf{u}_z$.

$$|\mathbf{u}_z| = 1$$

$$\cos \beta = \frac{\mathbf{u}'' \cdot \mathbf{u}_z}{|\mathbf{u}''| |\mathbf{u}_z|} = \frac{d}{l}$$

$$l^2 = a^2 + b^2 + c^2 = a^2 + d^2$$



- Comparing the coordinate-independent form of the cross-product

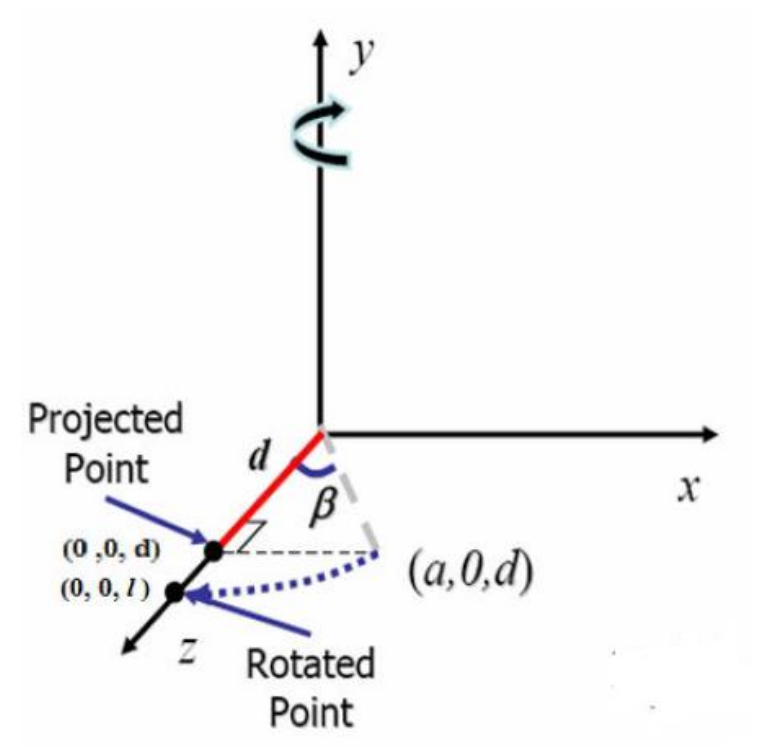
$$\mathbf{u}'' \times \mathbf{u}_z = \mathbf{u}_y |\mathbf{u}''| |\mathbf{u}_z| \sin \beta$$

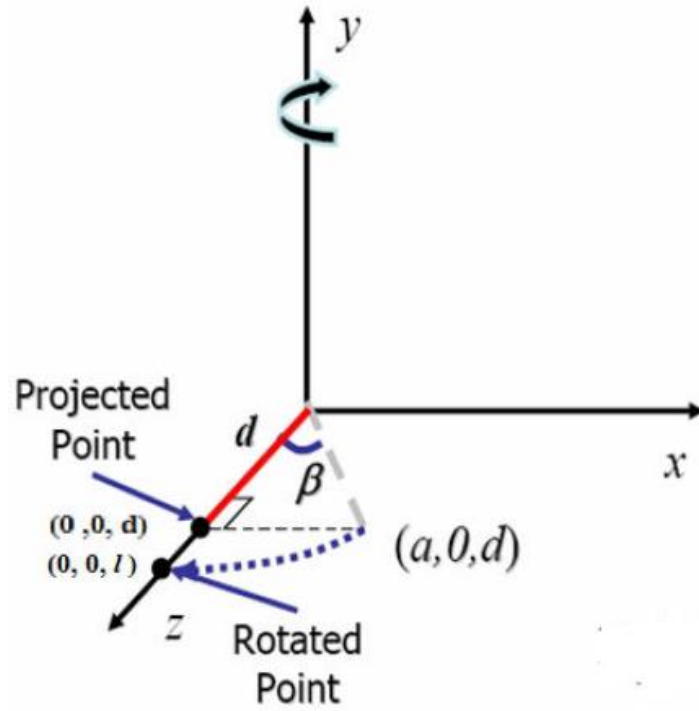
- with the Cartesian form

$$\mathbf{u}'' \times \mathbf{u}_z = \mathbf{u}_y \cdot (-a)$$

- $|\mathbf{u}''| = l$, we find that

$$\sin \beta = \frac{a}{l}$$





$$\sin \beta = \frac{a}{l}, \quad \cos \beta = \frac{d}{l}$$

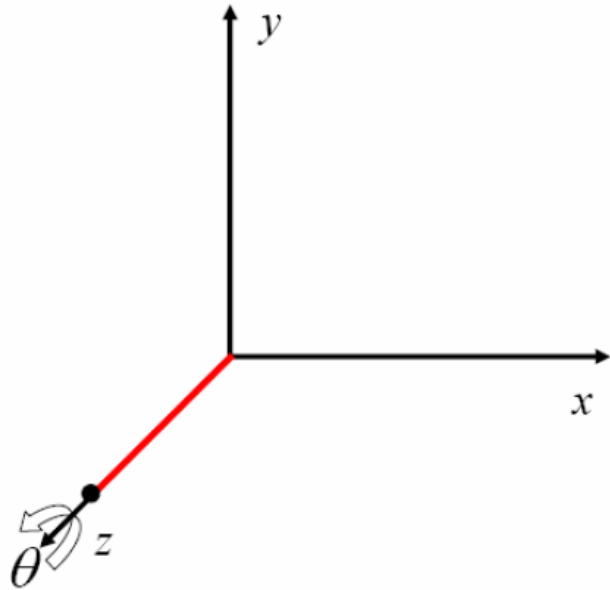
$$l^2 = a^2 + b^2 + c^2 = a^2 + d^2$$

$$\mathbf{R}_y(\beta) = \begin{bmatrix} \cos \beta & 0 & -\sin \beta & 0 \\ 0 & 1 & 0 & 0 \\ \sin \beta & 0 & \cos \beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} d/l & 0 & -a/l & 0 \\ 0 & 1 & 0 & 0 \\ a/l & 0 & d/l & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

U''

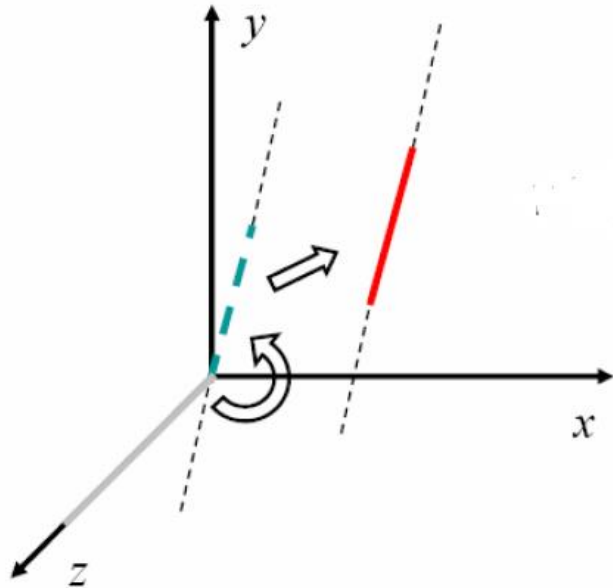
3. Perform the specified rotation about selected coordinate axis.

Here, rotation about z axis by the angle theta is done



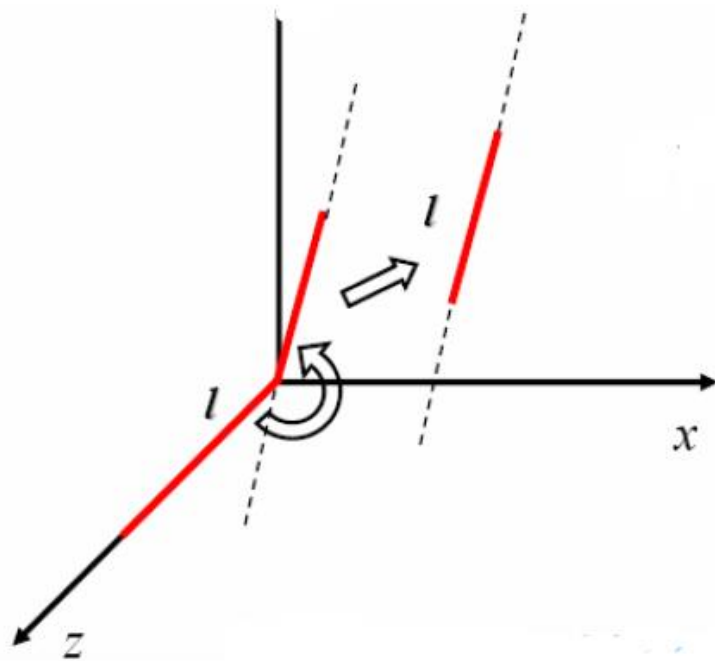
$$\mathbf{R}_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

4. Apply inverse rotation to bring the rotation axis back to its original orientation.



$$\mathbf{T}^{-1}\mathbf{R}_x^{-1}(\alpha)\mathbf{R}_y^{-1}(\beta)=\begin{bmatrix} 1 & 0 & 0 & x_1 \\ 0 & 1 & 0 & y_1 \\ 0 & 0 & 1 & z_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \alpha & \sin \alpha & 0 \\ 0 & -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
$$\begin{bmatrix} \cos \beta & 0 & \sin \beta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \beta & 0 & \cos \beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

5. Apply inverse translation to bring the rotation axis back to its original position.



$$\mathbf{T}^{-1}\mathbf{R}_x^{-1}(\alpha)\mathbf{R}_y^{-1}(\beta)=\begin{bmatrix} 1 & 0 & 0 & x_1 \\ 0 & 1 & 0 & y_1 \\ 0 & 0 & 1 & z_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \alpha & \sin \alpha & 0 \\ 0 & -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \cos \beta & 0 & \sin \beta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \beta & 0 & \cos \beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation about an Arbitrary Axis

- Composite Transformation

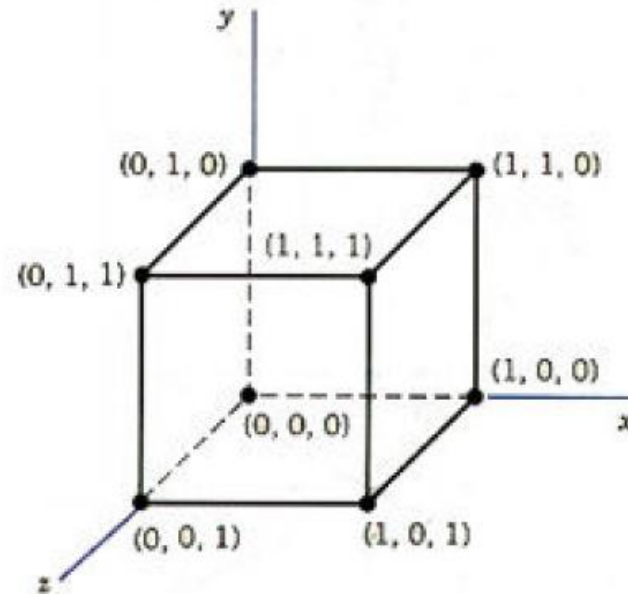
$$\mathbf{T}^{-1} \cdot \mathbf{R}_x^{-1}(\alpha) \cdot \mathbf{R}_y^{-1}(\beta) \cdot \mathbf{R}_z(\theta) \cdot \mathbf{R}_y(\beta) \cdot \mathbf{R}_x(\alpha) \cdot \mathbf{T}$$

- $M =$

$$M = T(x_1, y_1, z_1) \cdot R_x^{-1}(\alpha) \cdot R_y^{-1}(\beta) \cdot R_z(\theta) \cdot R_y(\beta) \cdot R_x(\alpha) \cdot T(-x_1, -y_1, -z_1)$$

Classwork

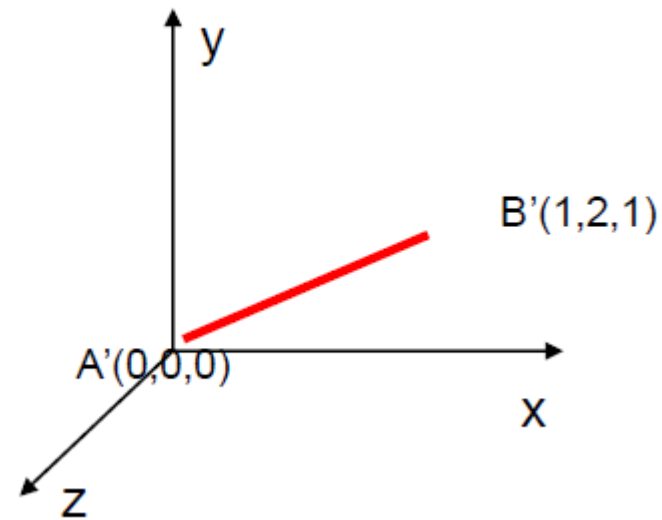
- ***Q. Find the new co-ordinates of a unit cube 90 degree rotated about an axis defined by its end points $A(2, 1, 0)$ and $B(3, 3, 1)$.***



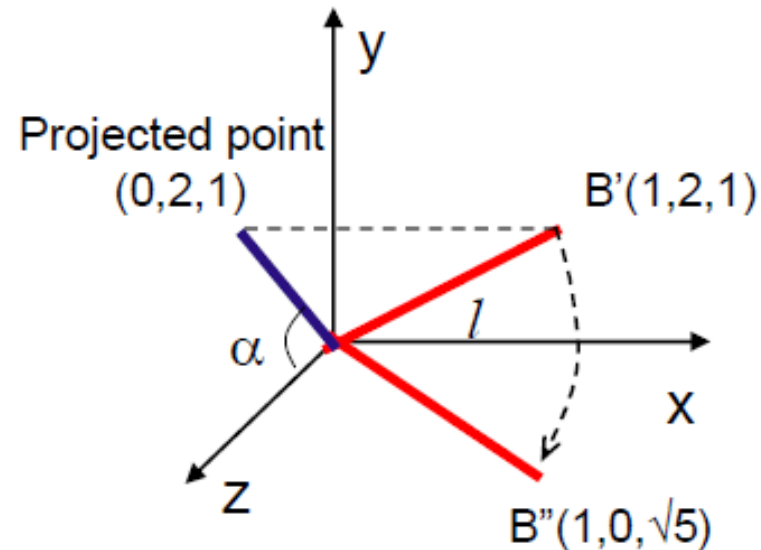
- Step1. Translating the point (A) to the origin,

$$T = \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$B - A = (3, 3, 1) - (2, 1, 0) = (1, 2, 1)$$



- Step 2a: Now, rotate the $A'B'$ about x-axis by angle α until it lies on xz-plane. Where,



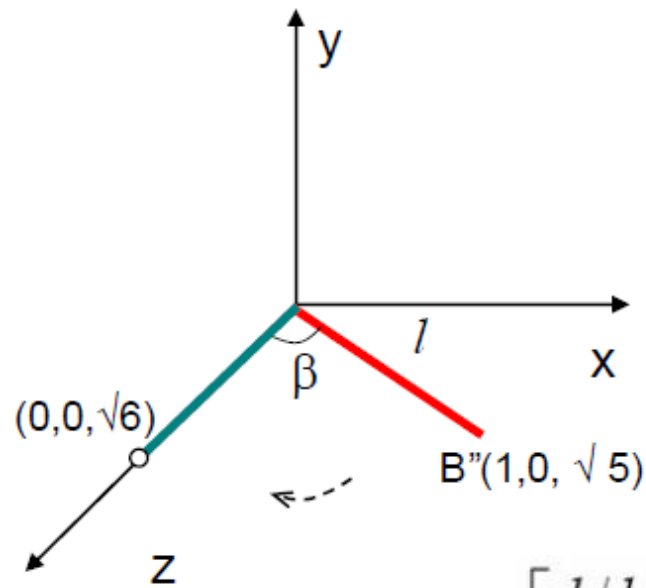
$$\sin \alpha = \frac{b}{\sqrt{b^2 + c^2}} = \frac{b}{d} \quad \sin \alpha = \frac{2}{\sqrt{2^2 + 1^2}} = \frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5}$$

$$\cos \alpha = \frac{c}{\sqrt{b^2 + c^2}} = \frac{c}{d} \quad \cos \alpha = \frac{1}{\sqrt{5}} = \frac{\sqrt{5}}{5}$$

$$l = \sqrt{1^2 + 2^2 + 1^2} = \sqrt{6}$$

$$R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha & 0 \\ 0 & \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c/d & -b/d & 0 \\ 0 & b/d & c/d & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/\sqrt{5} & -2/\sqrt{5} & 0 \\ 0 & 2/\sqrt{5} & 1/\sqrt{5} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Step 2b: Again, rotating $A'B'$ about y-axis by angle β until it coincides with z-axis.



$$\sin \beta = \frac{a}{l}, \quad \cos \beta = \frac{d}{l}$$

$$l^2 = a^2 + b^2 + c^2 = a^2 + d^2$$

$$l = \sqrt{1^2 + 2^2 + 1^2} = \sqrt{6}$$

$$R_y(\beta) = \begin{bmatrix} \cos \beta & 0 & \sin \beta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \beta & 0 & \cos \beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} d/l & 0 & -a/l & 0 \\ 0 & 1 & 0 & 0 \\ a/l & 0 & d/l & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \sqrt{5/6} & 0 & -1/\sqrt{6} & 0 \\ 0 & 1 & 0 & 0 \\ 1/\sqrt{6} & 0 & \sqrt{5/6} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Step 3: Rotating the unit cube 90 degree about z-axis.

$$R_z(90^\circ) = \begin{bmatrix} \cos 90^\circ & -\sin 90^\circ & 0 & 0 \\ \sin 90^\circ & \cos 90^\circ & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- The combined transformation rotation matrix about the arbitrary axis becomes

$$\mathbf{R}(\theta) = \mathbf{T}^{-1} \mathbf{R}_x^{-1}(\alpha) \mathbf{R}_y^{-1}(\beta) \mathbf{R}_z(90^\circ) \mathbf{R}_y(\beta) \mathbf{R}_x(\alpha) \mathbf{T}$$

$$\mathbf{R}(\theta) = \mathbf{T}^{-1} \mathbf{R}_x^{-1}(\alpha) \mathbf{R}_y^{-1}(\beta) \mathbf{R}_z(90^\circ) \mathbf{R}_y(\beta) \mathbf{R}_x(\alpha) \mathbf{T}$$

$$\mathbf{R}(\theta) = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{0}{\sqrt{5}} & \frac{2\sqrt{5}}{5} & 0 \\ 0 & -\frac{2\sqrt{5}}{5} & \frac{\sqrt{5}}{5} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{30}}{6} & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 1 & 0 & 0 \\ -\frac{\sqrt{6}}{6} & 0 & \frac{\sqrt{30}}{6} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{\sqrt{30}}{6} & 0 & -\frac{\sqrt{6}}{6} & 0 \\ \frac{0}{6} & 1 & \frac{0}{6} & 0 \\ \frac{\sqrt{6}}{6} & 0 & \frac{\sqrt{30}}{6} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{5} & -\frac{2\sqrt{5}}{5} & 0 \\ 0 & \frac{2\sqrt{5}}{5} & \frac{\sqrt{5}}{5} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.166 & -0.075 & 0.983 & 1.742 \\ 0.742 & 0.667 & 0.075 & -1.151 \\ -0.650 & 0.741 & 0.167 & 0.560 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Now, multiplying $R(\theta)$ by the matrix of original unit cube;

$$P' = R(\theta).P$$

$$P' = \begin{bmatrix} 0.116 & -0.075 & 0.983 & 1.742 \\ 0.742 & 0.667 & 0.075 & -1.151 \\ -0.650 & 0.741 & 0.167 & 0.560 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$P' = \begin{bmatrix} 2.650 & 1.667 & 1.834 & 2.816 & 2.7525 & 1.742 & 1.909 & 2.891 \\ -0.558 & -0.484 & 0.258 & 0.184 & -1.225 & -1.152 & -0.409 & -0.483 \\ 1.467 & 1.301 & 0.650 & 0.817 & 0.726 & 0.566 & -0.091 & 0.076 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

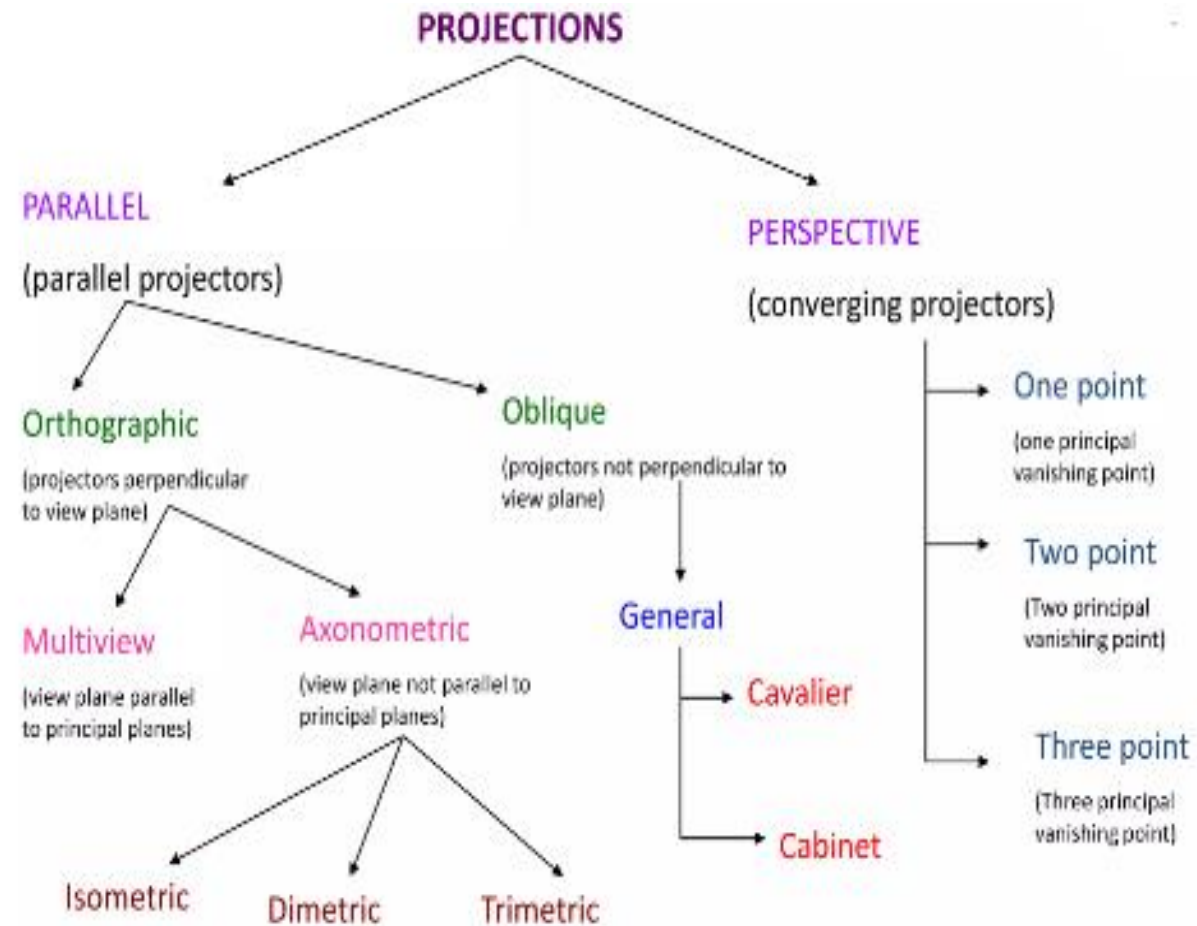
Classwork

- ***Q. A homogenous coordinate point $P(3, 2, 1)$ is translated in x, y, z direction by $-2, -2$ & -2 unit respectively followed by successive rotation of 60 about x - axis. Find the final position of homogenous coordinate.***
- ***Q. A cube of length 10 units having one of its corner at origin $(0, 0, 0)$ & three edges along principal axis. If the cube is to be rotated about z -axis by an angle of 450 in counter clockwise direction, calculate the new position of point***

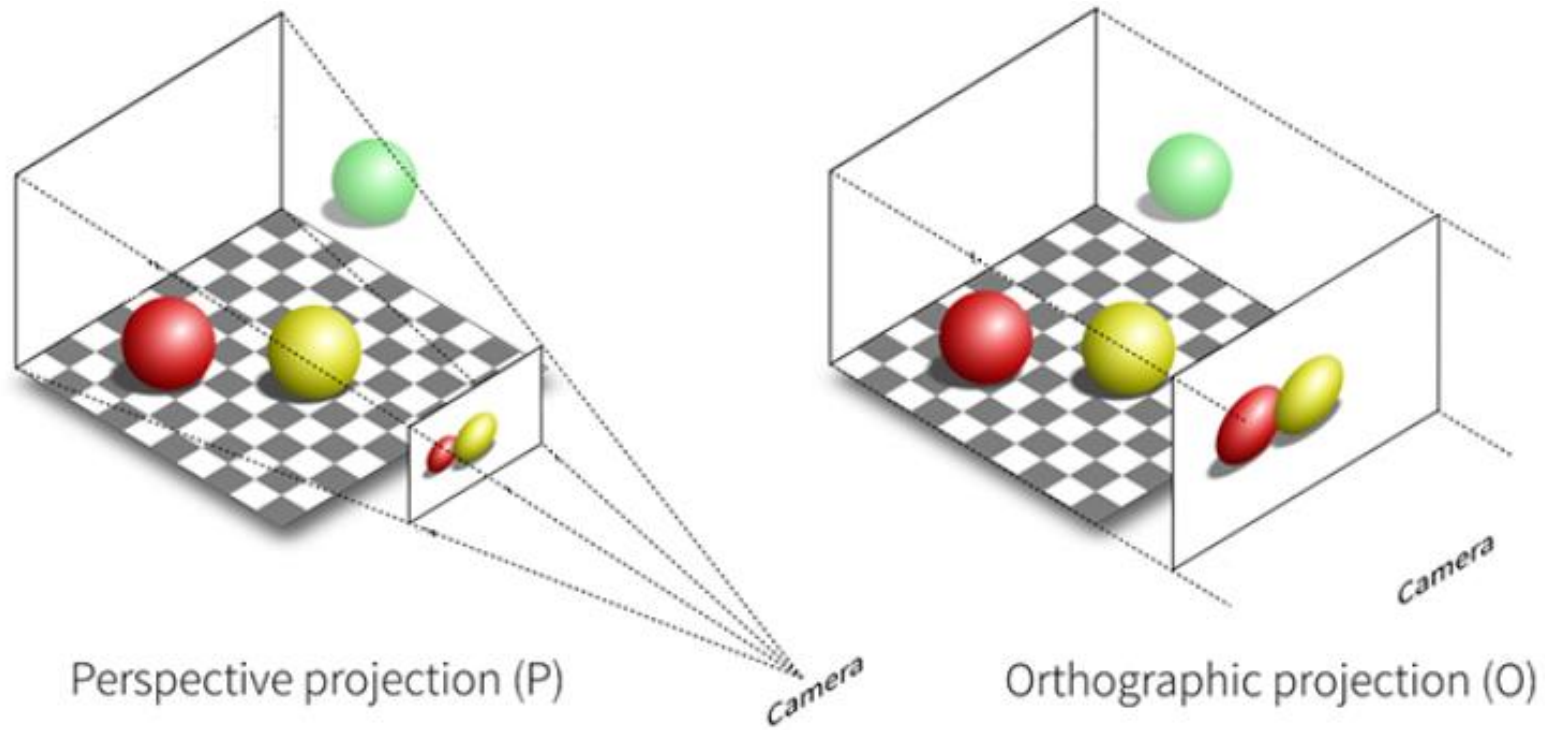
Projections

- Transformation that changes a point in n -dimensional coordinate system into a point in a coordinate system that has dimension less than n .
- Converts 3-D viewing co-ordinates to 2-D projection co-ordinates
- **TYPES OF PROJECTION**
 - Parallel Projection
 - Perspective Projection

• TYPES OF PROJECTION



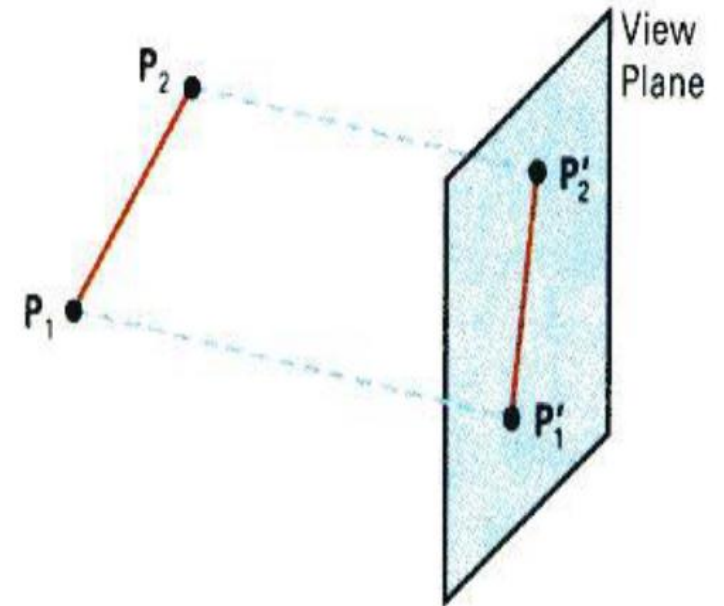
Projections



Projections

1. Parallel Projection

- Parallel Projection transforms object positions to the view plane along parallel lines
- A parallel projection preserves relative proportions of objects
- Accurate views of the various sides of an object are obtained with a parallel projection
- Can be used for measurements
 - Building plans
 - Manuals
- Cannot see what object really looks like because many surfaces hidden from view



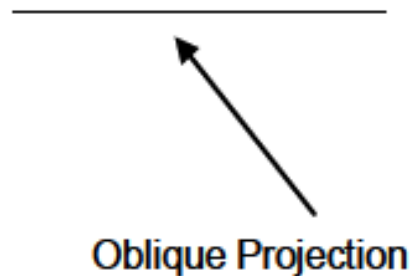
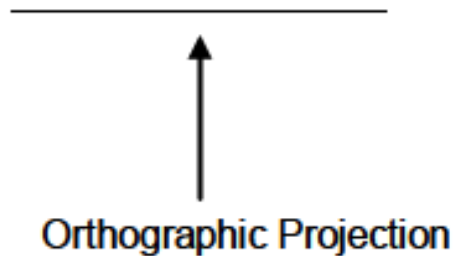
1. Parallel Projection Types

A. Orthographic Parallel Projection Types

- when the projection is perpendicular to the view plane

B. Oblique Parallel Projection Types

- when the projection is not perpendicular to the view plane



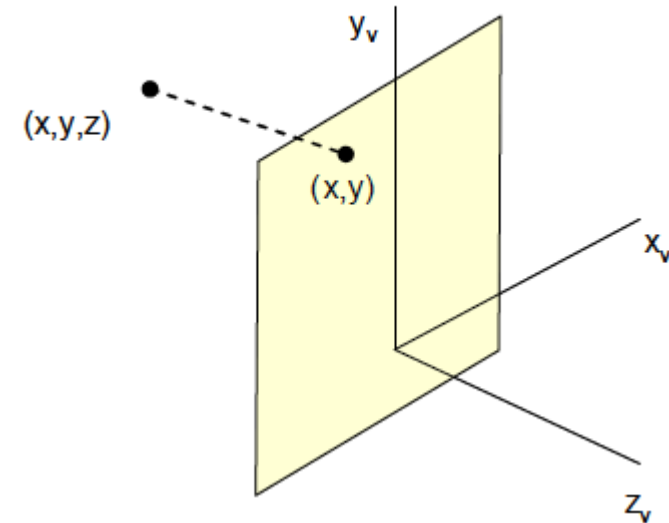
A. Orthographic Parallel Projection

- The transformation *equation* for orthographic projection is

$$X_p = x \quad Y_p = y \quad z = 0$$

- Z coordinate value is preserved for the depth
- In homogenous coordinate form,

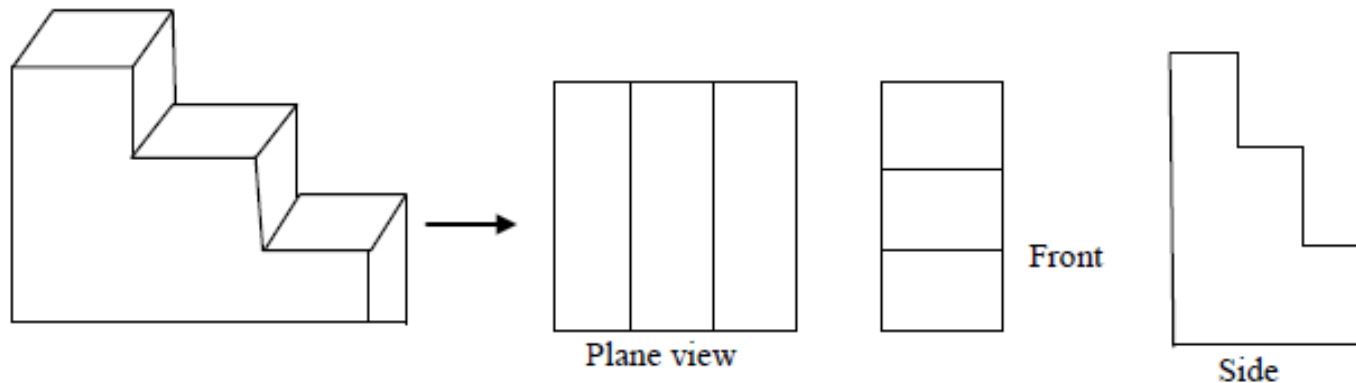
$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



A. Orthographic Parallel Projection Types

a. Multi View Projection

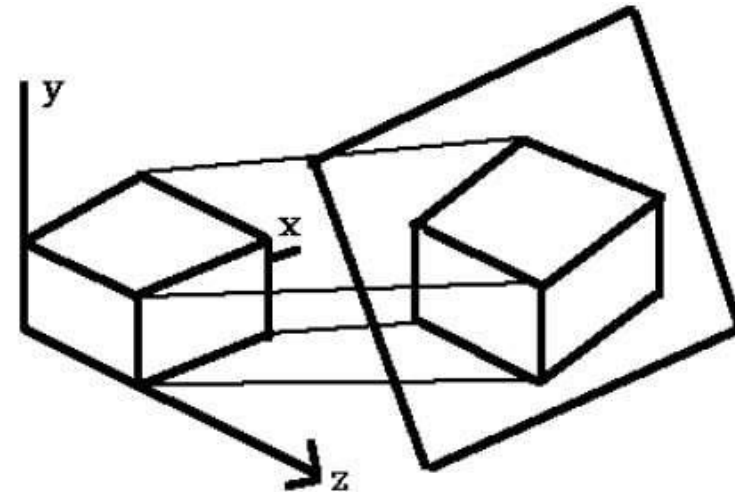
- Orthographic parallel projections are done by projecting points along parallel lines that are perpendicular to the projection plane
- Used to produce Front, Side and Top view of an object
- Front, side and rear orthographic are called elevations and the top orthographic view of object is known as plan view



b. Axonometric

Orthographic projections that show more than one side of an object are called axonometric orthographic projections. There are three axonometric projections are they are:

- i. **Isometric:** The most common axonometric projection is an **isometric projection** where the projection plane intersects each coordinate axis in the model coordinate system at an equal distance. In this projection parallelism of lines are preserved but angles are not preserved. The following figure shows isometric projection



- ii. **Dimetric projections:** In these two projectors have equal angles with respect to two principal axis.
- iii. **Trimetric projections:** The direction of projection makes unequal angle with their principal axis.

B. Oblique Parallel Projection

It is a kind of parallel projection where projecting rays emerges parallelly from the surface of the object and incident at an angle other than 90 degrees on the plane.

- Point (x, y, z) is projected to position (x_p, y_p) on the view plane.
- Orthographic projection coordinates on the plane are (x, y) . Oblique projection line from (x, y, z) to (x_p, y_p) makes an angle with the line on the projection plane that joins (x_p, y_p) and (x, y) . This line of length L is at an angle Φ with the horizontal direction in the projection plane.

- Expressing projection coordinates in terms of x, y, L and Φ as

$$x_p = x + L \cos \Phi$$

$$y_p = y + L \sin \Phi$$

L depends on the angle α and z coordinate of point to be projected

$$\tan \alpha = z/L$$

Thus,

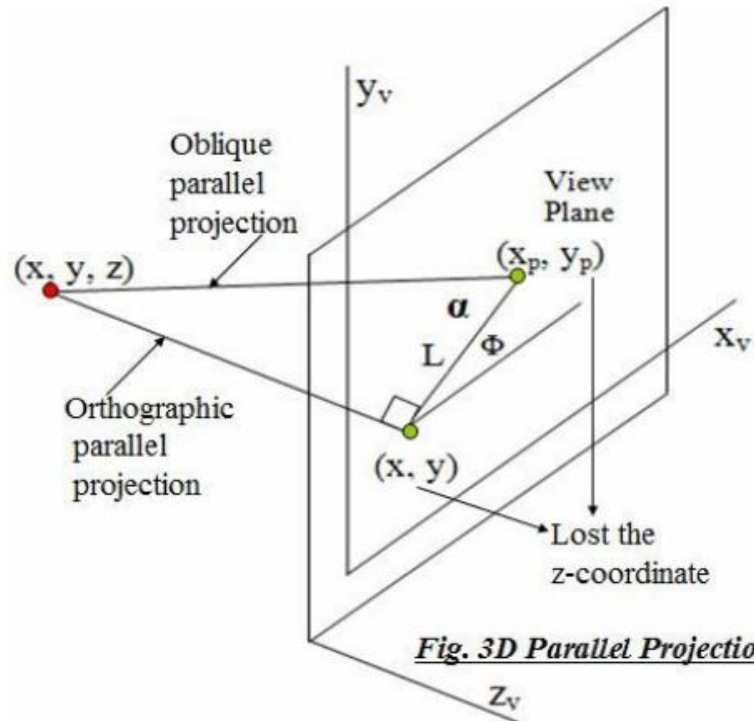
$$L = z / \tan \alpha$$

$$= z L_1, \text{ where } L_1 \text{ is the inverse of } \tan \alpha$$

So the oblique projection equations are:

$$x_p = x + z (L_1 \cos \Phi)$$

$$y_p = y + z (L_1 \sin \Phi)$$



- The transformation matrix for producing any parallel projection onto the $x_v y_v$ -plane can be written as:

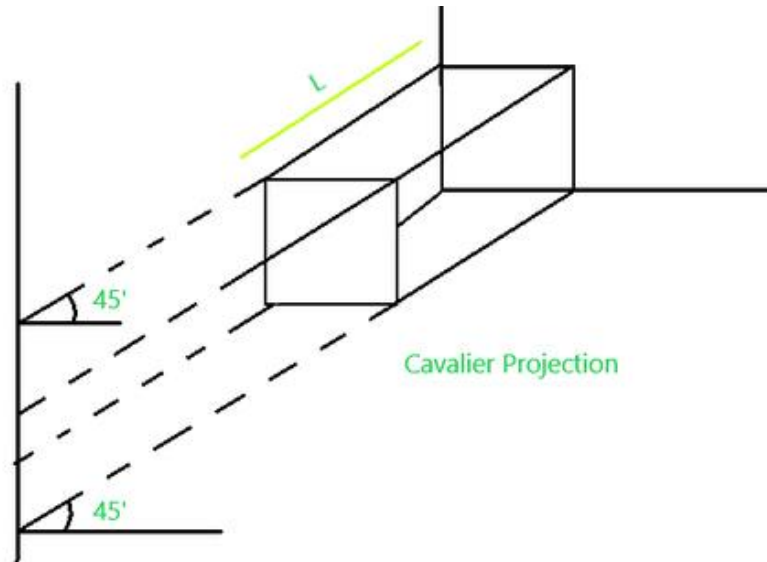
$$M_{\text{parallel}} = \begin{pmatrix} 1 & 0 & L_1 \cos \Phi & 0 \\ 0 & 1 & L_1 \sin \Phi & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Orthographic projection is obtained when $L_1 = 0$ (occurs at projection angle α of 90 degree) Oblique projection is obtained with non-zero values for L_1 .

B. Oblique Parallel Projection Types

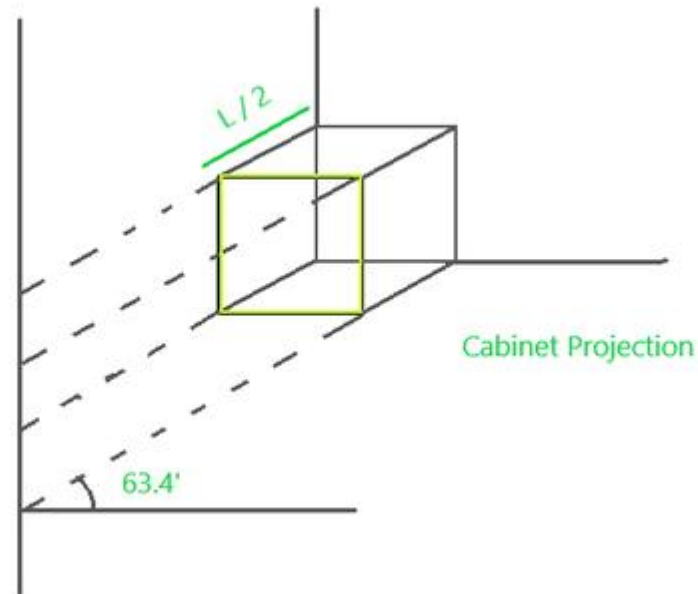
- It is of two kinds :

a) **Cavalier Projection** : It is a kind of oblique projection where the projecting lines emerge parallelly from the object surface and incident at 45° rather than 90° at the projecting plane. In this projection, the length of the reading axis is larger than the cabinet projection.



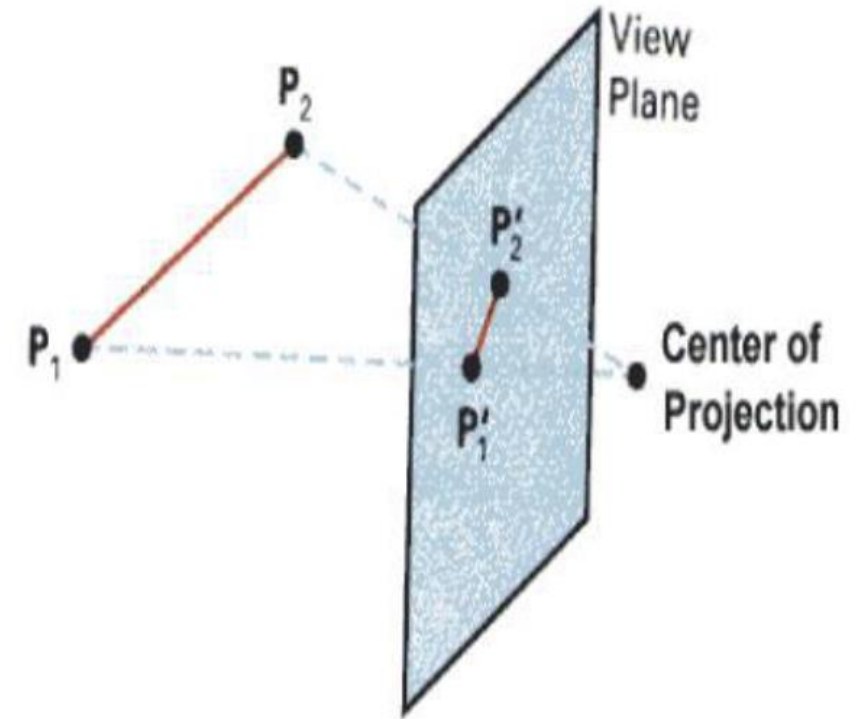
(b) Cabinet Projection :

It is similar to that cavalier projection but here the length of receding axes just half than the cavalier projection and the incident angle at the projecting plane is 63.4° rather 45° .

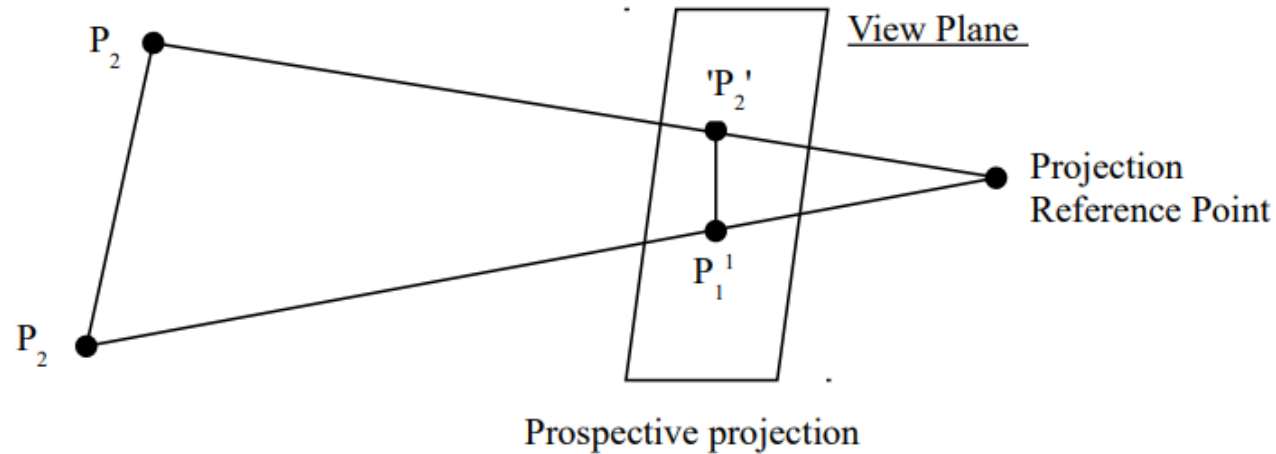


2.Perspective Projection

- ❑ Perspective Projection transforms object positions to the view plane while converging to a centre point of projection (projection reference point)
- ❑ Perspective projection produces realistic views but does not preserve relative proportions
- ❑ Projections of distant objects are smaller than the projections of objects of the same size that are closer to the projection plane

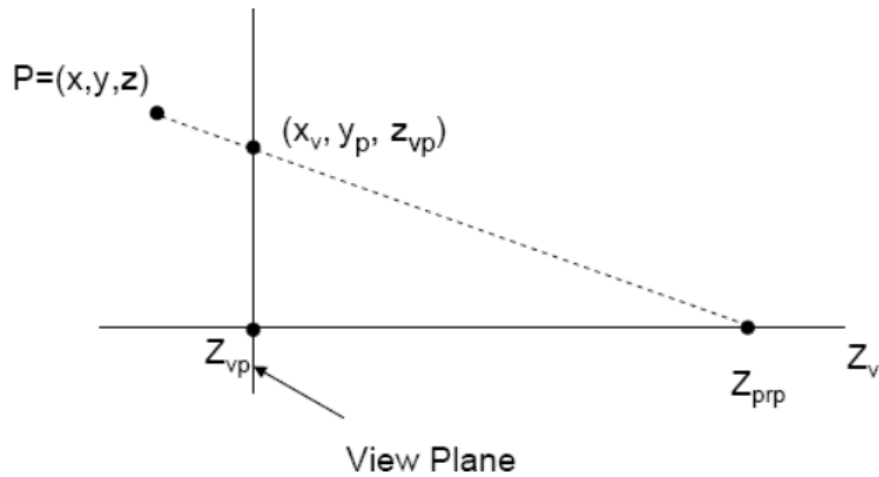


Perspective Projection



The projection is perspective if the distance is finite or has fixed vanishing point or the incoming rays converge at a point. Thus the viewing dimension of object is not exact i.e. seems large or small according as the movement of the view plane from the object.

The perspective projection perform the operation as the scaling (i.e. zoom in & out) but here the object size is only maximize to the size of real present object i.e. only size reduces not enlarge. This operation is possible here only the movement of the view plane towards or far from the object position.



- We can write equations describing co-ordinates positions parametric form as along this prospective projection line in

$$x' = x - xu$$

$$y' = y - yu$$

$$z' = z - (z - z_{prp})u$$

Where $u=0$ to 1 .

If $u = 0$, we are at position $P = (x, y, z)$. If $u = 1$, we have projection reference point $(0, 0, z_{prp})$.

On the view plane, $z' = z_{vp}$ then

$$u = \frac{z_{vp} - z}{z_{prp} - z}$$

Substituting these value in eq^n for x' , y' .

$$x_p = x - x\left(\frac{z_{vp} - z}{z_{prp} - z}\right) = x\left(\frac{z_{prp} - z_{vp}}{z_{prp} - z}\right) = x\left(\frac{dp}{z_{prp} - z}\right)$$

- Similarly,

$$y_p = y\left(\frac{z_{prp} - z_{vp}}{z - z_{prp}}\right) = y\left(\frac{dp}{z_{prp} - z}\right)$$

Where $dp = z_{prp} - z_{vp}$ is the distance of the view plane from projection reference point.

- Using 3-D homogeneous Co-ordinate representation, we can write prospective projection transformation matrix as

$$\begin{bmatrix} x_h \\ y_h \\ z_h \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & z_{vp}/dp & \frac{-z_{vp}}{z_{prp}/dp} \\ 0 & 0 & 1/dp & z_{prp}/dp \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Perspective Projection

- Obtain perspective projection co-ordinates for the pyramid with vertices of base(15, 15, 10), (20,20,10), (25,15,10), (20,10,10) and apex(20,15,20) given that $z_{prp} = 20$ and $z_{vp} = 0$.
 - (30,30,0)
 - (40,40,0)
 - (50,30,0)
 - (40,20,0)

Perspective Projection

Vanishing Point

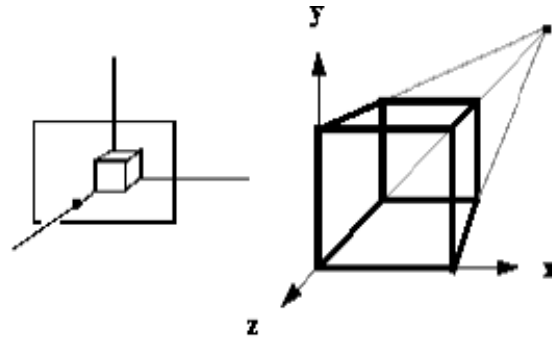
- Any set of parallel lines **that are not parallel** to view plane are projected as converging lines that appear to converge at a point called vanishing point

Principal Vanishing point

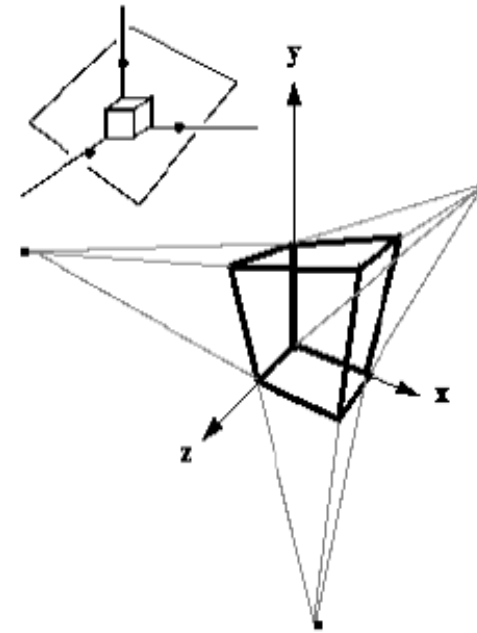
- If set of lines are parallel to one of the three principal axes, it is called principal vanishing point
- We can control the number of principal vanishing point to one, two or three with the orientation of projection plane and classify as **one, two or three point perspective projection**



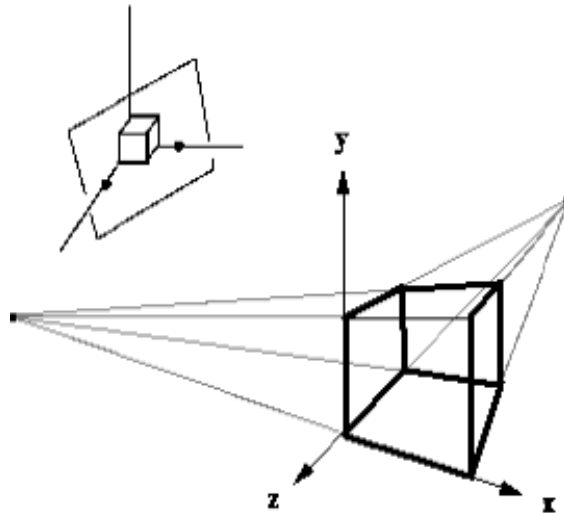
Vanishing points



One Point Perspective
(z-axis vanishing point)

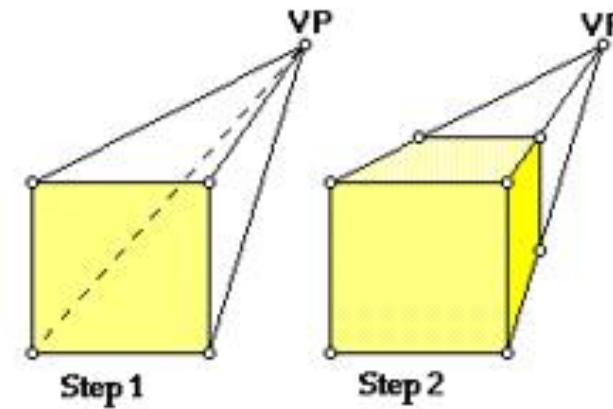


Three Point Perspective
(z, x, and y-axis
vanishing points)

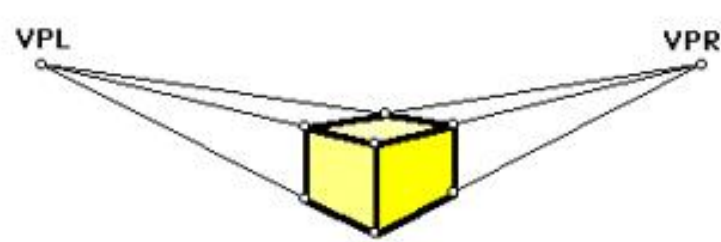
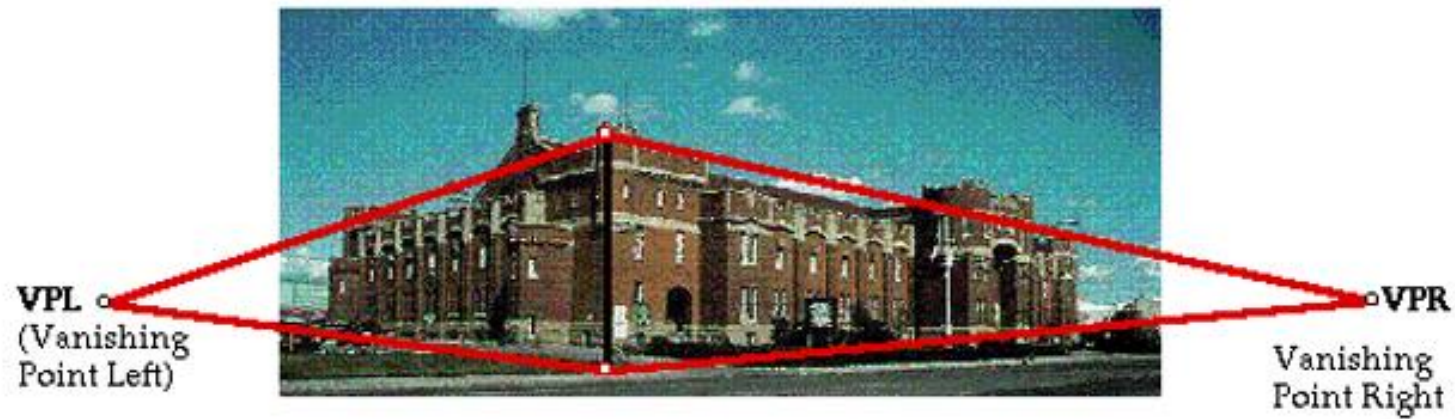


Two Point Perspective
z, and x-axis vanishing points

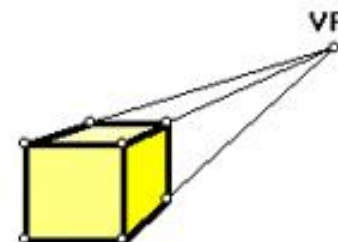
Vanishing Point



Vanishing Point



2-point perspective



1-point perspective

3-D Viewing

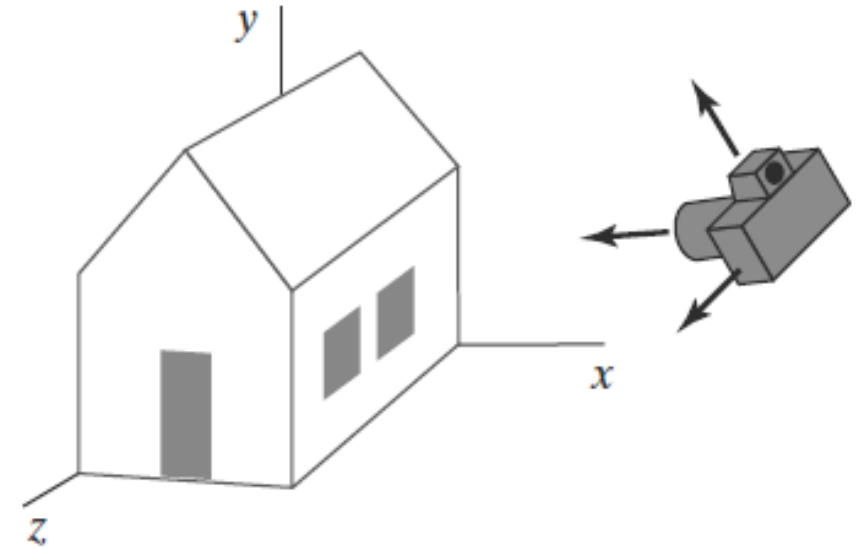
Viewing in 3D involves the following considerations:

- ❑ We can view an object from any spatial position, eg. in front of an object, behind the object, in the middle of a group of objects, inside an object, etc.
- ❑ 3D descriptions of objects must be projected onto the flat viewing surface of the output device
- ❑ The clipping boundaries enclose a volume of space
- ❑ 3-D scene generation process analogous to taking photographs with camera

❑ For a snapshot, we need to position the camera at a particular point in space and then need to decide camera orientation

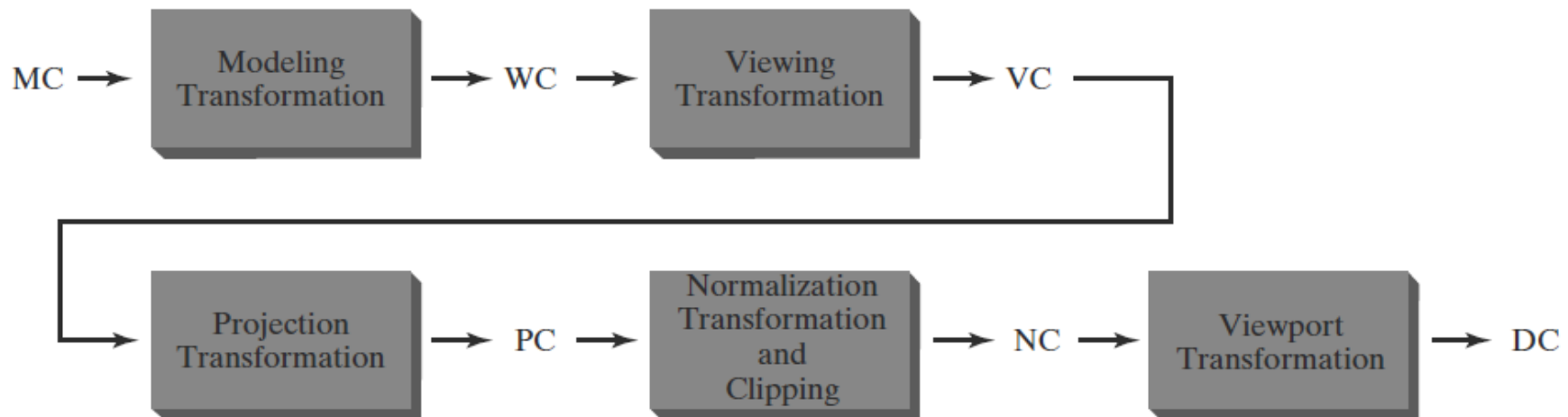
❑ Finally when we snap the shutter, the seen is cropped to the size of window of the camera and the light from the visible surfaces is projected into the camera film

❑ Once the scene has been modelled, world coordinate position are converted to viewing coordinates



3-D Viewing Pipeline

- Viewing in 3D Viewing involves the following steps:
- Modeling coordinates (MC) to world coordinates (WC) to viewing coordinates (VC) to projection coordinates (PC) to normalize coordinates (NC) and, ultimately, to device coordinates (DC)

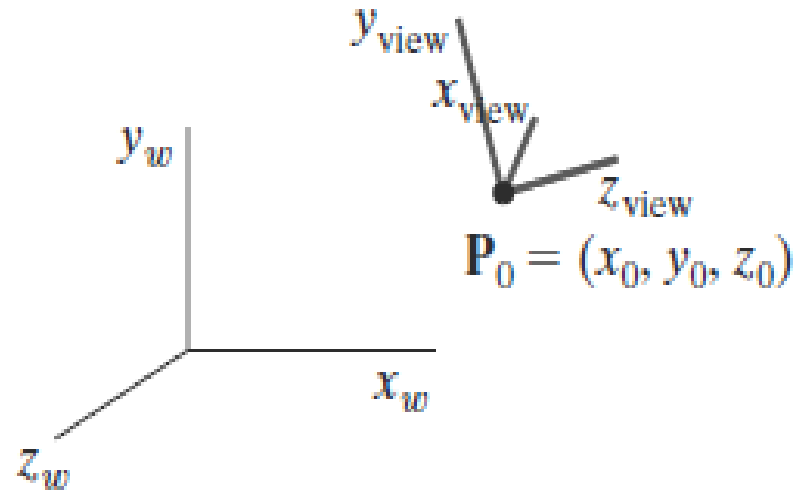


- Each object has its own dimension and modelling coordinate. The **Modelling Transformation** can be done by 3D transformations
- After modeling transformation, the objects are placed in a scene or a **world coordinate system**
- The viewing-coordinate system is used in graphics packages as a reference for specifying the observer **viewing position** and the **position of the projection plane** analogous to camera film plane. This is viewing transformation where we get **viewing coordinates**.
- Projection operations convert the viewing-coordinate description (3D) to coordinate positions on the projection plane (2D view plane) (Usually combined with clipping, visual-surface identification, and surface rendering). This **projection transformation** creates projection coordinates.

- ❑ Workstation transformation(**Normalization transformation & clipping and view port transformation**) maps the coordinate positions on the projection plane to the output device
- ❑ The clipping window is positioned on a selected view plane, and scenes are clipped against an enclosing volume of space, which is defined by a set of *clipping planes*
- ❑ The viewing position, view plane, clipping window, and clipping planes are all specified within the viewing-coordinate reference frame
- ❑ The final step is to map viewport coordinates to **device coordinates** within a selected display window

Transformation From World To Viewing Coordinates

- world-coordinate position $\mathbf{P}_0 = (x_0, y_0, z_0)$. for the viewing origin, which is called the **view point** or **viewing position**.



- Figure illustrates the positioning of a three-dimensional viewing-coordinate frame within a world system

- Conversion of object descriptions from world to viewing coordinates is equivalent to a transformation that superimposes the viewing reference frame onto the world frame using the basic geometric **translate rotate** operations:
 1. Translate the view reference point to the origin of the world coordinate system
 2. Apply rotations to align the x_v , y_v , and z_v axes (viewing coordinate system) with the world x_w , y_w , z_w axes, respectively.

- The viewing-coordinate origin is at world position $P_0 = (x_0, y_0, z_0)$. Therefore, the matrix for translating the viewing origin to the world origin is

$$T = \begin{bmatrix} 1 & 0 & 0 & -x_0 \\ 0 & 1 & 0 & -y_0 \\ 0 & 0 & 1 & -z_0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- For the rotation transformation, we can use the unit vectors u , v , and n to form the composite rotation matrix that superimposes the viewing axes onto the world frame

$$R = \begin{bmatrix} u_x & u_y & u_z & 0 \\ v_x & v_y & v_z & 0 \\ n_x & n_y & n_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- The coordinate transformation matrix is then obtained as the product of the preceding translation and rotation matrices:

$$\begin{aligned}\mathbf{M}_{WC, VC} &= \mathbf{R} \cdot \mathbf{T} \\ &= \begin{bmatrix} u_x & u_y & u_z & -\mathbf{u} \cdot \mathbf{P}_0 \\ v_x & v_y & v_z & -\mathbf{v} \cdot \mathbf{P}_0 \\ n_x & n_y & n_z & -\mathbf{n} \cdot \mathbf{P}_0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\end{aligned}$$

- Matrix transfers world-coordinate object descriptions to the viewing reference frame.

THANK YOU